

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_41c8dc8a-8a4\agent-a060c32a49a9792f4.jsonl`  
- SHA-256: `f3771d716590975d7a4dd8e09450a6453e52b2e768c293138811a16feb58f6b0`  
- Source modified: `2026-07-02T02:54:34+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_41c8dc8a-8a4`  
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T02:47:11.981Z)

CONTEXT (read carefully):

- You are one auditor in a multi-agent tech-debt/critical-fix audit of the khelsutra-guru multi-repo workspace at C:/Users/avidu/Projects/khelsutra-guru. Today is 2026-07-02. All git repos were just pulled to current origin master/main.
- Repos: badminton-highlight-indexer (the ENGINE, Python, ~1500 tests, org Khelsutra), khelsutra (SPA web/ + marketing site/, TS/React, org avidullu), sports-data-collector, rally-annotator, sports-obsreport (small Python repos). Plus two NON-git dirs: "Annotation Setup", khelsutra-screencast.
- HARD RULES: READ-ONLY. Do NOT modify/create/delete any file, do NOT create branches/worktrees, do NOT pip install, do NOT run the full test suite (CI is trusted on master). You MAY run read-only git and gh commands (gh is authenticated), ruff/mypy IF already installed, and fast read-only scripts.
- EVERY claim must cite file:line (or a gh URL). A claim without file:line evidence is a hypothesis - mark it as such or drop it. The codebase drifts between sessions: re-verify anything you believe from docs against actual code.
- OWNER-GATED areas (flag as owner_gated=true, still report): alpha/serving go-live + Cloudflare dashboard work, Studio P10/P11 (corpus promotion + train/reeval), product/strategy/licensing decisions, real GPU/cloud spend, counsel ToS/DPA, repo rename off "badminton".
- Known-open items from project memory (VERIFY current state, don't trust blindly): engine PR #390 (D1 split main.py, open since 06-26) · audit doc docs/CODE_CLEANUP_AUDIT_2026-06-24.md remaining items D1, D13 (ByteTrack->trackers), Phase-3 (B3/B4/B6/B7/B8/D5/D6/D8-D10/O3/O4) · docs/DECISION_SNAPSHOT_REGRESSION.md next=P1 · docs/GOLDEN_REGRESSION_FIXTURES.md backlog M4/P0-rename/O1/O2 · engine issues #340 (Gemini re-bill), #332 (NSFW preflight), #349 (CPU-bound decode), #384 (CI single-source) · khelsutra issues #97-#114 UX/alpha backlog · DO droplet RETIRED 2026-06-25 (docs may still reference it; api.khelsutra.guru CNAME/CF-Access cleanup was pending).
- Categorize each finding: critical-fix (latent bug/correctness/data-loss/cost-leak), security, tech-debt (structure/duplication/dead code), test-gap, ci-infra, doc-rot, hygiene.
- Severity: P0 = latent bug or cost/data/security risk reachable in normal use; P1 = high-value debt blocking velocity or a stale-open thread; P2 = worthwhile cleanup; P3 = nice-to-have (report max 3 of these).
- Be selective: max ~12 findings, each genuinely worth acting on. Note tracked_in when an issue/PR/doc already tracks it (tracked is NOT done - still report). Your final message is data for an orchestrator, not prose for a human.

YOUR DIMENSION: full audit of rally-annotator. Repo: C:/Users/avidu/Projects/khelsutra-guru/rally-annotator.

Small repo (annotation tool; HEAD = #25 README sync after a multi-agent audit). Audit: code quality, tests + CI (gh run list), the 2 open issues (#26 'More than just a CSV?', #22 i18n native review) - verify current state; packaging/run instructions accuracy (can a fresh user actually run it per README?); overlap/duplication with the 'Annotation Setup' non-git dir at 'C:/Users/avidu/Projects/khelsutra-guru/Annotation Setup' (peek at it read-only - is it a duplicate/predecessor of this repo?).

2. user (2026-07-02T02:47:18.945Z)

```
3d604bc docs: sync READMEs with current state (multi-agent audit fixes) (#25)
f5ceefc docs(readme): tick live-test ? v1.7.1 verified in real VLC (incl. Indic rendering) (#24)
a0592f6 docs(readme): refresh stale roadmap + tests lines (#23)
ac245ce feat(vlc): complete localization ? status messages + HELP guide (v1.7.1) (#21)
245bf54 feat(vlc): localize dialog chrome into 7 languages (v1.7) (#20)
total 66
drwxr-xr-x 1 avidu 197609      0 Jun 22 15:03 .
drwxr-xr-x 1 avidu 197609      0 Jun 29 20:32 ..
drwxr-xr-x 1 avidu 197609      0 Jul  2 08:17 .git
-rw-r--r-- 1 avidu 197609    369 Jun 13 21:50 .gitattributes
drwxr-xr-x 1 avidu 197609      0 Jun 22 03:33 .github
-rw-r--r-- 1 avidu 197609    123 Jun  7 06:58 .gitignore
-rw-r--r-- 1 avidu 197609   13899 Jun 22 03:33 CHANGELOG.md
-rw-r--r-- 1 avidu 197609    4386 Jun 22 03:33 CONTRIBUTING.md
-rw-r--r-- 1 avidu 197609    1066 Jun  7 06:57 LICENSE
-rw-r--r-- 1 avidu 197609   13058 Jun 22 15:03 README.md
drwxr-xr-x 1 avidu 197609      0 Jun 22 03:33 docs
drwxr-xr-x 1 avidu 197609      0 Jun 22 03:33 test
drwxr-xr-x 1 avidu 197609      0 Jun 22 03:33 vlc
drwxr-xr-x 1 avidu 197609      0 Jun 22 15:03 web
```

3. user (2026-07-02T02:47:19.162Z)

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total 28
drwxr-xr-x 1 avidu 197609      0 Jun 23 09:53 .
drwxr-xr-x 1 avidu 197609      0 Jun 29 20:32 ..
drwxr-xr-x 1 avidu 197609      0 Jun 23 09:53 Archive
drwxr-xr-x 1 avidu 197609      0 Jun 22 15:12 Golden Labelled
drwxr-xr-x 1 avidu 197609      0 Jun 18 12:26 Trajectories
-rw-r--r-- 1 avidu 197609    2508 Jun  6 14:16 golden_labels_badminton.csv
-rw-r--r-- 1 avidu 197609    1377 Jun  6 13:14 rally_review.csv
```

4. user (2026-07-02T02:47:25.696Z)

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./.gitattributes
./.github/pull_request_template.md
./.github/workflows/ci.yml
./.github/workflows/web-ci.yml
./.gitignore
./CHANGELOG.md
./CONTRIBUTING.md
./docs/CSV_FORMAT.md
./docs/ENDING_REASONS.md
./docs/LOCALIZATION.md
./LICENSE
./README.md
./test/dialog_layout.snapshot
./test/dialog_test.lua
./test/README.md
./vlc/rally_annotator.lua
./web/.gitignore
./web/DESIGN.md
./web/e2e/extension.spec.ts
./web/e2e/fixtures/index.html
./web/e2e/fixtures/sample.webm
./web/e2e/serve.mjs
./web/entrypoints/background.ts
./web/entrypoints/content.ts
./web/package-lock.json
./web/package.json
./web/playwright.config.ts
./web/README.md
./web/src/i18n/index.ts
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./web/src/i18n/locales/da/common.json
./web/src/i18n/locales/en/common.json
./web/src/i18n/locales/es/common.json
./web/src/i18n/locales/hi/common.json
./web/src/i18n/locales/id/common.json
./web/src/i18n/locales/kn/common.json
./web/src/i18n/locales/te/common.json
./web/src/messages.ts
./web/src/persist/store.ts
./web/src/state/annotator.ts
./web/src/state/csv.ts
./web/src/ui/panel.ts
./web/src/version.ts
./web/src/video/directVideo.ts
./web/test/annotator.test.ts
./web/test/csv.test.ts
./web/test/directVideo.test.ts
./web/test/i18n.test.ts
./web/test/panel.smoke.test.ts
./web/test/store.test.ts
./web/TESTING.md
./web/tsconfig.json
./web/vitest.config.ts
./web/wxt.config.ts

```

5. user (2026-07-02T02:47:25.824Z)

```

1      # Rally Annotator
2
3      [![VLC dialog suite](https://github.com/Khelsutra/rally-
annotator/actions/workflows/ci.yml/badge.svg)](https://github.com/Khelsutra/rally-
annotator/actions/workflows/ci.yml)
4      [![web extension CI](https://github.com/Khelsutra/rally-annotator/actions/workflows/web-
ci.yml/badge.svg)](https://github.com/Khelsutra/rally-annotator/actions/workflows/web-ci.yml)
5
6      > **Part of [Khelsutra](https://khelsutra.guru) ? _?Every rally, indexed. The dead time,
gone.?_**
7      > Khelsutra turns a full badminton match into a clean index of rallies with the dead
time stripped out ?
8      > a highlight reel you can watch and download. **Rally Annotator is its open labeling
tool**: the
9      > ground-truth rally CSVs produced here are what the Khelsutra pipeline trains and
evaluates its
10     > rally-segmentation models against. This repo is the public, open-source piece of that
stack.
11
12     Hand-label rally **start/end + point-stop reason** while you watch a match, for **net-
separated racquet
13     sports** ? badminton, tennis, table tennis, pickleball, padel. Two front-ends, **same
CSV**:
14
15     - ? **VLC plugin** ? label **local video files** inside VLC (this README); the dialog is
**localized**
16     (English, Hindi, Kannada, Telugu, Spanish, Danish, Indonesian) via a Language
selector.
17     - ? **Browser extension** _(new)_ ? label **web video, including YouTube**, right in the
page;
18     **localized into English, Hindi, Kannada, Telugu, Spanish, Danish & Indonesian**. See
**[web/](web/)**.
19
20     You watch the video in VLC, pause/scrub freely, and click **Mark START** / **Mark END**
(with a
21     reason). It writes one CSV row per rally ? clean ground-truth labels for
training/evaluating rally
22     (point) segmentation models, or just for cutting highlights.

```

```

23
24     ``
25     rally_number,start_time,end_time,ending_reason,sport,shots_count
26     1,8.800,11.500,winner,badminton,9
27     2,24.389,46.589,unforced_error,badminton,21
28     ``
29
30     Times are decimal seconds; `shots_count` is an optional rally shot/stroke count
31     (blank if you skip it).
32     See [docs/CSV_FORMAT.md](docs/CSV_FORMAT.md).
33
34     ## Generating rally annotations ? pick your front-end
35
36     | You're labeling? | Use | Output |
37     |---|---|---|
38     | a local video file | the VLC plugin (install + use below) |
39     | <video>.rallies.csv` next to the file |
40     | a web video / YouTube (`youtube.com/watch`) | the browser extension ?
41     [web/](web/) | the full CSV downloaded on each save |
42
43     Both write the identical CSV schema, so labels from either path feed the same
44     Khelsutra pipeline. The
45     VLC flow is documented below; the browser extension has its own [README](web/README.md),
46     a
47     [design doc with locked decisions](web/DESIGN.md), and a [testing
48     strategy](web/TESTING.md). Either way the
49     workflow is the same: Mark START ? Mark END ? pick the reason ? Save, one CSV row
50     per rally.
51
52     ## Why
53
54     Labeling rally boundaries is the slow, expensive prerequisite for any rally-detection
55     model. Most tools
56     make you transcribe timestamps by hand. This lets a rater do it inside the player they
57     already use,
58     pausing to the exact frame before each mark ? turning "watch a match" into "produce
59     golden labels."
60
61     ## Install
62
63     1. Copy `vlc/rally_annotator.lua` into your VLC Lua extensions folder:
64     - Windows: `%APPDATA%\vlc\lua\extensions\`
65     - macOS: `~/Library/Application Support/org.videolan.vlc/lua/extensions/`
66     - Linux: `~/.local/share/vlc/lua/extensions/`
67     (create the `extensions` folder if it doesn't exist)
68     2. In VLC: Tools ? Plugins and extensions ? Reload extensions (or just restart VLC).
69     3. Enable it from the View menu ? Rally Annotator. A small dialog opens and
70     stays open while you watch.
71
72     Requires VLC 3.0.x. (VLC 4.0 changed the Lua input/listener API; targeting 3.x for
73     now.)
74
75     ## Use
76
77     Playback is built in ? the Back 5s · Play / Pause · Fwd 5s row drives the VLC
78     player from the
79     annotation window, so you can pause, label, and resume without ever switching to the
80     main VLC window. Play / Pause
81     is a single toggle (it resumes when paused, pauses when playing).
82
83     1. Pick the Sport (it stays set across rallies).
84     2. When a rally begins, click Mark START ? it snapshots the current playback time
85     into the Start field
86     (pause/scrub first for frame accuracy; you can also edit the field by hand).
87     3. When the rally ends, click Mark END (fills the End field). This arms the
88     rally; nothing is written yet.
89     4. Choose the Ending reason (sits between Mark END and Save Rally; defaults
90     to `unknown`), optionally

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71         type a **Number of shots** (the rally's shot/stroke count ? leave blank to skip),
then click
72         **Save Rally** to write one CSV row. The reason **resets to `unknown` after every
save** (and the shots field
73         clears too) ? so neither silently reuses the previous rally's value, and you can pick
the same reason on
74         consecutive rallies. Use the **Next rally #** field to resume/insert numbering
anywhere.
75     5. **Recent rallies** list: select a row, then **Edit selected** (loads it back into the
fields to fix
76         start/end/reason/sport ? **Save changes**) or **Delete selected**. **Undo last**
removes the most recent row ?
77         the button shows which one, e.g. `Undo last (#7)`, and the status panel shows that
row's details ? or clears an
78         in-progress mark, or cancels an edit.
79     6. **Help** button ? usage + an [ending-reason decision guide](docs/ENDING_REASONS.md),
rendered inside the dialog.
80
81     **Output:** `<video-stem>.rallies.csv` next to the video (falls back to your home dir if
the path can't be
82     resolved). Re-opening the extension **continues** rally numbering from the existing file
? no duplicate IDs.
83
84     **Resuming a half-finished video:** labels are saved to that CSV as you go. Open the
**same video** and enable the
85     extension and it **reloads your existing rallies** (shown in the Recent list) and
continues numbering ? so you can
86     stop and pick up later. Easiest flow: **open the video first, then enable the
extension**. If you enable it **before**
87     the video is loaded, that's fine too ? the moment you click **Mark START** (the video is
playing by then) the tool
88     switches to that video's own `<video-stem>.rallies.csv` and loads any rallies already
saved for it, so nothing is
89     written to the home-dir fallback by mistake. If you switch videos with the dialog
already open, click **Refresh** to
90     point at the current video and load its rallies. (Playback **position** isn't restored ?
scrub to where you stopped.)
91
92     > **Where's the help / "About" text?** The empty **"About ?lua" box under **Tools ?
**Plugins and extensions ?
93     > Add-ons** is VLC's own add-on info dialog; its body ("Lua script") is a constant baked
into VLC and **can't be
94     > set by an extension**. Use the **Help** button inside the Rally Annotator dialog
instead. (The descriptor's
95     > description does show in the **Active Extensions* tab's "More information" dialog,
and the **Active Extensions*
96     > list shows the title with its version (it tracks the plugin's `VERSION`, currently
**`Rally Annotator v1.7.1`**), so you can confirm which build is loaded.)
97
98     ## Sports & taxonomy
99     Net-separated racquet sports share a forced/unforced-error point-stop taxonomy, so one
tool covers them all:
100
101     | Sport | Notes |
102     |---|---|
103     | badminton | the reference sport |
104     | tennis | service_fault = fault/double-fault; let supported |
105     | table_tennis | let (net serve) supported; fast cadence |
106     | pickleball | service_fault for faults; treat "let" per local rules |
107     | padel | net-separated; rally boundaries as in tennis |
108
109     `ending_reason` ? `{unknown, winner, forced_error, unforced_error, service_fault, let,
other}` ? a shared
110     vocabulary across these sports. Keep to these values for clean downstream aggregation.
111

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112     ### What the ending reasons mean
113     `unknown` is the default (the field resets to it after every save); every other
reason except `winner` is
114     charged to the side that lost the rally.
115
116     | Reason | When the rally ended because? |
117     |---|---|
118     | `unknown` | default ? not classified yet. A save with no reason picked records
`unknown` (never the previous rally's reason); set a specific reason when you can. |
119     | `winner` | the last shot landed in and went unreturned (opponent couldn't reach it
/ only waved). A clean ace counts here. |
120     | `forced_error` | the loser missed (out or into the net) while under pressure ?
stretched, rushed, jammed, handling the opponent's pace/spin/depth. |
121     | `unforced_error` | the loser missed a routine shot they had time and position
to make, with little or no pressure. |
122     | `service_fault` | the point ended on the serve ? into the net, out of the service
box, illegal action/foot fault, or a double fault. |
123     | `let` | the rally is replayed under the rules, no point scored (e.g. a
tennis/table-tennis serve net-cord that's otherwise good, outside interference). |
124     | `other` | none of the above ? occluded footage, injury/retirement, penalty, hindrance.
Use sparingly. |
125
126     The cases people ask about: a ball/shuttle landing out is the hitter's error ?
`forced_error` only if they
127     were under pressure, otherwise `unforced_error` (it's never automatically forced,
and never a winner for the
128     opponent). Into the net during a rally is the same; a serve into the net is
`service_fault`. When unsure
129     forced-vs-unforced, default to `unforced_error`. See the full decision guide with
examples and per-sport rules
130     in [docs/ENDING_REASONS.md](docs/ENDING_REASONS.md) (also available via the in-
plugin Help button).
131
132     ## Roadmap
133     - [x] Live-tested the v1.7.1 dialog in real VLC ? loads and renders correctly, the
new Language
134         selector rebuilds the whole dialog, and Kannada / Telugu / Hindi render crisply
in VLC's Qt with
135         the system Noto/Nirmala fonts on the tested machine (bundling OFL Noto for
guaranteed offline / any-machine
136         rendering is a tracked follow-up ? see
[docs/LOCALIZATION.md](docs/LOCALIZATION.md)). The mark?save logic is
137         covered by the headless suite (88 assertions); per-sport marking spot-checks on
real clips still welcome.
138     - [x] Localization into Khelsutra's languages + Telugu (en/hi/kn/te/es/da/id):
both front-ends
139         fully localized ? the browser extension and the entire VLC dialog (chrome,
status messages, and
140         the in-dialog ending-reason guide). Machine drafts pending native review ? see
[docs/LOCALIZATION.md](docs/LOCALIZATION.md).
141
142     - [ ] Optional per-sport reason presets / hotkeys.
143     - [x] Browser front-end for web video incl. YouTube ? Chrome MV3 extension emitting
the same CSV
144         ([web/](web/)). Firefox/Safari build legs and embedded-iframe
(`youtube.com/embed`) support are
145         documented follow-ups (see [web/DESIGN.md](web/DESIGN.md)).
146     - [ ] A `python-vlc` + Tk/Qt app with global keyboard shortcuts (S/E/1?6/U) for power
users / remote raters.
147     - [x] Optional `shots_count` column (v1.6). Further extra columns (e.g. server/receiver)
still TBD behind a toggle.
148
149     ## Tests
150     The dialog logic has a headless test suite that stubs VLC's `vlc` API (widgets +
playback), loads the real
151     extension, drives its callbacks, and asserts the results ? run with Lua 5.1 (the

```

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interpreter VLC 3.x embeds):
152
153     ``bash
154     lua5.1 test/dialog_test.lua          # exit 0 = pass; covers reason reset, shots,
155     ``                                     numbering, undo, help toggle, Play/Pause, i18n key-parity + language switch
156
157     It covers all the logic but not VLC's GUI rendering or real playback (there's no
158     headless VLC UI automation) ? those
159     are checked by loading in VLC with `vlc -vv --file-logging` (must scan with no `Error
160     loading` line) plus a manual
161     click-through. See [test/README.md](test/README.md).
162
163     ## About Khelsutra
164     [Khelsutra](https://khelsutra.guru) turns a full badminton match into a clean
165     index of rallies with
166     the dead time stripped out ? a highlight reel you can watch and download, plus a
167     summary. It's an
168     invite-only alpha (Bengaluru) aimed at academies and players, with privacy built
169     in (your footage
170     stays yours; minors are protected) and a roadmap toward shot detection, deeper match
171     analysis, per-match
172     history, and an AI assistant coach ? expanding to more sports and languages over
173     time.
174
175     Rally Annotator is the open-source labeling tool of that stack. The rally CSVs you
176     produce here are the
177     ground truth its rally-segmentation models learn from ? so better labels directly mean
178     better auto-generated
179     reels. _?Every rally, indexed. The dead time, gone._
180
181     ## Contributing
182     Issues and PRs welcome ? live test reports per sport/OS, browser-extension features, and
183     UX fixes. Every
184     change is gated by tests (which is what lets us merge outside PRs with confidence): keep
185     the VLC suite
186     (`lua5.1 test/dialog_test.lua`) and the web suite (`cd web && npm test`) green and add a
187     case for new logic.
188     See [CONTRIBUTING.md](CONTRIBUTING.md) for the full dev setup, the rigorous merge
189     bar each PR must
190     pass, and how to add tests for a change.
191
192     ## License
193     MIT ? see [LICENSE](LICENSE).
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```

6. user (2026-07-02T02:47:36.115Z)

```

26      OPEN      More than just a CSV?                2026-06-23T04:21:12Z
22      OPEN      i18n: native-speaker review of machine-draft catalogs (hi/kn/te/es/da/id)
2026-06-22T03:52:14Z
---PRS---
---RUNS---
completed      success docs: sync READMEs with current state (multi-agent audit fixes) (#25)
web-extension CI      main      push      27933033995      1m15s      2026-06-22T06:07:10Z
completed      success docs: sync READMEs with current state (multi-agent audit fixes) (#25)
tests main      push      27933033987      18s      2026-06-22T06:07:10Z
completed      success docs: sync READMEs with current state (audit fixes)      tests
docs/readme-sync      pull_request      27932962182      20s      2026-06-22T06:05:26Z
completed      success docs: sync READMEs with current state (audit fixes)      web-extension CI
docs/readme-sync      pull_request      27932962160      1m13s      2026-06-22T06:05:26Z
completed      success docs: sync READMEs with current state (multi-agent audit fixes) web-
extension CI      docs/readme-sync      push      27932960879      1m21s      2026-06-22T06:05:24Z
completed      success docs(readme): tick live-test ? v1.7.1 verified in real VLC (incl. Ind?
tests main      push      27930372144      28s      2026-06-22T04:51:52Z

```

```

completed      success docs(readme): tick live-test ? v1.7.1 verified live in VLC      tests
docs/live-test-verified pull_request      27930352015      21s      2026-06-22T04:51:12Z
completed      success docs(readme): refresh stale roadmap + tests lines (#23) tests      main
push      27928483135      15s      2026-06-22T03:55:24Z
completed      success docs(readme): refresh stale roadmap + tests lines      tests
docs/readme-roadmap-refresh pull_request      27928461163      20s      2026-06-22T03:54:43Z
completed      success feat(vlc): complete localization ? status messages + HELP guide (v1.7?
tests main      push      27915515172      16s      2026-06-21T19:47:47Z
completed      success feat(vlc): complete localization ? status messages + HELP guide (v1.7.1)
tests feat/vlc-i18n-complete pull_request      27915496760      17s      2026-06-21T19:47:01Z
completed      success feat(vlc): localize dialog chrome into 7 languages (v1.7) (#20) tests
main      push      27915079995      17s      2026-06-21T19:30:14Z
completed      success feat(vlc): localize dialog chrome into 7 languages (v1.7)      tests
feat/vlc-i18n      pull_request      27915067753      14s      2026-06-21T19:29:48Z
completed      success feat(web): localize panel into kn/hi/es/da/id/te + per-locale screenc?
tests main      push      27914405861      16s      2026-06-21T19:02:57Z
completed      success feat(web): localize panel into kn/hi/es/da/id/te + per-locale screenc?
web-extension CI      main      push      27914405856      1m11s      2026-06-21T19:02:57Z

```

7. user (2026-07-02T02:47:36.148Z)

```

1      name: web-extension CI
2
3      # Proof-and-validation pipeline for the browser extension (web/). Runs on changes
4      # under web/. Unit + coverage (enforced thresholds), typecheck, build, then a REAL
5      # Chromium E2E that loads the built extension and drives it on a real <video> ? the
6      # recorded screencast and coverage are uploaded as artifacts.
7      on:
8        push:
9          paths:
10             - "web/**"
11             - ".github/workflows/web-ci.yml"
12        pull_request:
13          paths:
14             - "web/**"
15             - ".github/workflows/web-ci.yml"
16
17      jobs:
18        test:
19          runs-on: ubuntu-latest
20          defaults:
21            run:
22              working-directory: web
23          steps:
24            - uses: actions/checkout@v4
25
26            - uses: actions/setup-node@v4
27              with:
28                node-version: 22
29                cache: npm
30                cache-dependency-path: web/package-lock.json
31
32            - run: npm ci
33
34            - name: Typecheck
35              run: npm run typecheck
36
37            - name: Unit tests + coverage (enforced thresholds)
38              run: npm run coverage
39
40            - name: Build extension (chrome-mv3)
41              run: npm run build
42
43            - name: Install Playwright Chromium
44              run: npx playwright install --with-deps chromium

```



```

45
46     - name: Install Indic/Noto fonts (so hi/kn/te render in the screencasts)
47       run: sudo apt-get update && sudo apt-get install -y fonts-noto-core fonts-indic
48
49     - name: E2E in real Chromium ? per-locale screencasts
50       run: xvfb-run -a npm run e2e
51       env:
52         CI: "true"
53
54     - name: Upload E2E screencast + report
55       uses: actions/upload-artifact@v4
56       if: always()
57       with:
58         name: chromium-e2e-screencast
59         path: |
60           web/e2e-results/screencasts
61           web/e2e-results/videos
62           web/e2e-report
63         if-no-files-found: warn
64
65     - name: Upload coverage report
66       uses: actions/upload-artifact@v4
67       if: always()
68       with:
69         name: coverage
70         path: web/coverage
71         if-no-files-found: warn
72

```

8. user (2026-07-02T02:47:36.148Z)

```

1     name: tests
2
3     # Runs the headless dialog suite on every push to main and every PR. VLC can't be
4     # driven headlessly (Qt GUI, no offscreen render, no CLI to auto-open a Lua extension),
5     # so instead of pixels we exercise the extension's logic AND diff its full widget grid +
6     # window title against a committed golden -- the deterministic stand-in for a
screenshot.
7     on:
8       push:
9         branches: [main]
10      pull_request:
11
12     jobs:
13       dialog-suite:
14         name: Lua 5.1 headless dialog suite
15         runs-on: ubuntu-latest
16         steps:
17           - uses: actions/checkout@v4
18
19           # Lua 5.1 is the exact interpreter VLC 3.x embeds, so test semantics match the
player.
20           - name: Install Lua 5.1
21             run: sudo apt-get update && sudo apt-get install -y lua5.1
22
23           # Syntax gate: the extension and the test must parse. A scan-time syntax error is
how
24           # the extension silently failed to load in v1.3.1, so we fail fast and loudly
here.
25           - name: Syntax-check the extension and tests
26             run: luac5.1 -p vlc/rally_annotator.lua test/dialog_test.lua
27
28           # Logic (mark/save/edit/undo, reason reset, numbering, help, playback) + the
layout
29           # snapshot (widget grid + title vs test/dialog_layout.snapshot). Exit 0 = all

```

```
pass.  
30         - name: Run dialog tests (logic + layout snapshot)  
31         run: lua5.1 test/dialog_test.lua  
32
```

9. user (2026-07-02T02:47:47.645Z)

```
title: More than just a CSV?  
state: OPEN  
author: avidullu (Avi)  
labels:  
comments: 0  
assignees:  
projects:  
milestone:  
number: 26  
--
```

Currently the csv generated is completely unattributable to the video file which was used to create it. Is there something we can do to record the path of the file, name, some critical metadata eg. MD5 hash and store it in the csv itself which helps attribute the rally csv to the original video in case the videos and rallies need to be stored separately?

====ISSUE22====

```
title: i18n: native-speaker review of machine-draft catalogs (hi/kn/te/es/da/id)  
state: OPEN  
author: avidullu (Avi)  
labels:  
comments: 0  
assignees:  
projects:  
milestone:  
number: 22  
--
```

Native-speaker review of the machine-draft translation catalogs

The localization (en ? **hi, kn, te, es, da, id**) shipped as **machine drafts** across both front-ends, deliberately marked ``pending-native-review``. Per the standing guardrail (mirrored from Khelsutra: ***AI drafts are inference-only, never the source of truth***), **native-speaker review** is the quality gate before any public launch. **This issue tracks that review ? one sign-off per language.**

What to review

- **Web extension catalogs:** ``web/src/i18n/locales/{hi,kn,te,es,da,id}/common.json`` (each stamped ``_meta.status=machine-draft``).
- **VLC plugin catalog:** the ``STRINGS`` table in ``vlc/rally_annotator.lua`` (same keys; includes the long in-dialog **ending-reason HELP** guide ? the part where racquet-sports terminology matters most).
- Focus: natural UI phrasing, correct racquet-sports terms (rally, shuttle, serve, fault, let, winner/forced/unforced error), and that the **ending-reason guide** reads correctly.

Per-language sign-off

- [] Hindi (hi)
- [] Kannada (kn)
- [] Telugu (te)
- [] Spanish (es) ? note CLDR ``es`` is 3-form plural, though our strings are phrased plural-free
- [] Danish (da)
- [] Indonesian (id)

How (no code changes needed)

Edit the ``common.json`` values (web) / ``STRINGS`` values (VLC) in place; **keys and placeholders** must stay identical ? the **key-parity CI gate** (web ``npm test``, VLC ``lua5.1 test/dialog_test.lua``) enforces structure, and reason/sport **values written to the CSV** stay canonical English regardless of the labels. When a language is reviewed, drop its ``_meta.status`` to ``reviewed``.

Design + locked decisions: `docs/LOCALIZATION.md`. Related (Khelsutra site Telugu):
avidullu/khelsutra#72.

10. user (2026-07-02T02:47:47.924Z)

```
1497 vlc/rally_annotator.lua
437 test/dialog_test.lua
306 web/src/state/annotator.ts
79 web/src/state/csv.ts
403 web/src/ui/panel.ts
50 web/entrypoints/content.ts
64 web/src/persist/store.ts
2836 total
```

11. user (2026-07-02T02:47:53.485Z)

```
1      // CSV serialize/parse for rally rows.
2      //
3      // This is a faithful port of the VLC extension's save_all() / load_rows()
4      // (vlc/rally_annotator.lua) and is intentionally BYTE-COMPATIBLE with it, so the
5      // CSV this browser extension produces ingests into the same pipeline
6      // (sports-data-collector -> badminton-highlight-indexer). See docs/CSV_FORMAT.md.
7      //
8      // Columns: rally_number,start_time,end_time,ending_reason,sport,shots_count
9      //   - times are decimal seconds, 3 dp (Lua "%.3f" <-> JS toFixed(3))
10     //   - shots_count is OPTIONAL: blank when not recorded
11     //   - any columns beyond shots_count (col 7+) are preserved verbatim on rewrite,
12     //     so richer downstream metadata is never dropped.
13
14     export interface RallyRow {
15       n: number; // rally_number (integer, 1-based, monotonic)
16       s: number; // start_time, decimal seconds
17       e: number; // end_time, decimal seconds (> s)
18       reason: string; // ending_reason
19       sport: string;
20       shots: string | null; // shots_count token verbatim (e.g. "9"); null when blank
21       extra: string | null; // columns 7+ joined verbatim; null when none
22     }
23
24     export const HEADER =
25       "rally_number,start_time,end_time,ending_reason,sport,shots_count\n";
26
27     // Mirror Lua tonumber(): trims, "" -> nil, non-numeric -> nil. (NB: JS Number(""))
28     // is 0 and Number(" 1 ") is 1, so we must special-case empty after trimming.)
29     function luaToNumber(x: string | undefined): number | null {
30       if (x == null) return null;
31       const t = x.trim();
32       if (t === "") return null;
33       const v = Number(t);
34       return Number.isNaN(v) ? null : v;
35     }
36
37     // Rewrite the whole CSV from rows (header once) ? mirrors save_all(). The browser
38     // has no atomic tmp+rename; durability is handled by the IndexedDB live store
39     // (see persist layer), with this serialized form as the downloadable artifact.
40     export function serializeRows(rows: RallyRow[]): string {
41       let out = HEADER;
42       for (const r of rows) {
43         const shots = r.shots != null ? r.shots : "";
44         let line =
45           `${Math.trunc(r.n)},${r.s.toFixed(3)},${r.e.toFixed(3)},${r.reason},${r.sport},${shots}`;
46         if (r.extra != null && r.extra !== "") line += "," + r.extra;
47         out += line + "\n";
48       }
49       return out;
50     }
```

```

49     }
50
51     // Forgiving parse ? mirrors load_rows(): any line whose first field is an integer
52     // is a rally row; header and blank lines are skipped; column 6 is the optional
53     // shots_count (blank/missing in older CSVs); columns 7+ are kept verbatim.
54     export function parseRows(text: string): RallyRow[] {
55         const rows: RallyRow[] = [];
56         for (const raw of text.split(/\n/)) {
57             const line = raw.replace(/\r$/, "");
58             if (!/^\S/.test(line)) continue; // skip blank lines
59             const parts = line.split(",");
60             const nRaw = luaToNumber(parts[0]);
61             if (nRaw == null || nRaw !== Math.floor(nRaw)) continue; // not a row (e.g. header)
62             let shots: string | null = parts[5] ?? null;
63             if (shots === "") shots = null; // blank => not recorded
64             // NB: the LOAD default for a missing/blank reason is "other" (distinct from the
65             // in-app SAVE default of "unknown"); preserved here exactly as in load_rows().
66             const reason = parts[3] != null && parts[3] !== "" ? parts[3] : "other";
67             const extra = parts.length > 6 ? parts.slice(6).join(",") : null;
68             rows.push({
69                 n: Math.floor(nRaw),
70                 s: luaToNumber(parts[1]) ?? 0,
71                 e: luaToNumber(parts[2]) ?? 0,
72                 reason,
73                 sport: parts[4] ?? "",
74                 shots,
75                 extra,
76             });
77         }
78         return rows;
79     }
80

```

12. user (2026-07-02T02:47:53.675Z)

```

1     // Annotator state machine ? a faithful, provider-agnostic port of the pure logic
2     // in vlc/rally_annotator.lua (the button callbacks + helpers, minus VLC widget I/O).
3     //
4     // Localizable by design: operations and button labels return message KEYS + params
5     // (a `Msg`), never prose ? the UI translates them via the i18n shim, so the state
6     // machine stays pure and i18n-free (keys-not-prose; see docs/LOCALIZATION.md). The
7     // reason/sport values it stores are canonical English (written verbatim to the CSV);
8     // only their DISPLAY labels are translated, by the UI.
9
10    import { RallyRow, serializeRows } from "./csv";
11
12    export const SPORTS = [
13        "badminton",
14        "tennis",
15        "table_tennis",
16        "pickleball",
17        "padel",
18    ] as const;
19
20    export const REASON_DEFAULT = "unknown";
21    export const REASONS = [
22        "winner",
23        "forced_error",
24        "unforced_error",
25        "service_fault",
26        "let",
27        "other",
28    ] as const;
29    // Dropdown order: the savable default ("unknown") first so it shows selected.
30    export const REASON_OPTIONS = [REASON_DEFAULT, ...REASONS] as const;

```

```

31
32 export type Mode = "new" | "edit";
33
34 export interface PersistResult {
35     ok: boolean;
36     err?: string;
37 }
38 export type PersistFn = (rows: RallyRow[]) => PersistResult;
39
40 const okPersist: PersistFn = () => ({ ok: true });
41
42 // Mirror get_field_num(): strip ALL whitespace (not just ends), "" -> null,
43 // non-numeric -> null.
44 function fieldNum(text: string): number | null {
45     const t = text.replace(/\s+/g, "");
46     if (t === "") return null;
47     const v = Number(t);
48     return Number.isNaN(v) ? null : v;
49 }
50
51 // A translatable message: a catalog key + optional interpolation params. The UI renders
52 // it with the i18n shim. `reason`/`sport` params hold canonical values; the UI maps
53 // them
54 // to translated labels at render time (the CSV always keeps the canonical value).
55 export interface Msg {
56     key: string;
57     params?: Record<string, string | number>;
58 }
59 export interface OpResult extends Msg {
60     ok: boolean;
61 }
62
63 export interface AnnotatorOpts {
64     rows?: RallyRow[];
65     persist?: PersistFn;
66 }
67
68 export class Annotator {
69     rows: RallyRow[];
70     mode: Mode = "new";
71     editIndex: number | null = null;
72
73     // Form fields mirror the editable widgets (kept as strings, like the text inputs).
74     startField = "";
75     endField = "";
76     shotsField = "";
77     nextField = "";
78     reason: string = REASON_DEFAULT; // non-sticky
79     sport: string = SPORTS[0]; // sticky
80
81     private persist: PersistFn;
82
83     constructor(opts: AnnotatorOpts = {}) {
84         this.persist = opts.persist ?? okPersist;
85         this.rows = opts.rows ?? [];
86         this.refreshNextField();
87     }
88
89     // ---- pure helpers (ported 1:1) ----
90
91     nextRallyNumber(): number {
92         let maxn = 0;
93         for (const r of this.rows) if (r.n > maxn) maxn = r.n;
94         return maxn + 1;
95     }

```

```

95
96     indexOfRally(n: number): number {
97         return this.rows.findIndex((r) => r.n === n);
98     }
99
100    // Smallest integer >= start not already used (auto-advance skips occupied numbers).
101    nextFreeFrom(start: number): number {
102        let n = start;
103        while (this.indexOfRally(n) !== -1) n++;
104        return n;
105    }
106
107    // The number the next NEW rally will get (the "Next rally #" override, else max+1).
108    plannedNextNumber(): number {
109        const v = fieldNum(this.nextField);
110        if (v !== null && v >= 1) return Math.floor(v);
111        return this.nextRallyNumber();
112    }
113
114    // Optional shots_count: null for blank/negative/non-numeric; else floor. NB: 0 is
115    // VALID (the Lua treats only nil/negative as "no count", and 0 is truthy there).
116    getShots(): number | null {
117        const v = fieldNum(this.shotsField);
118        if (v == null || v < 0) return null;
119        return Math.floor(v);
120    }
121
122    // Fresh rally fully marked (START + END) but not yet saved -> unsaved work exists.
123    isArmed(): boolean {
124        return (
125            this.mode === "new" &&
126            fieldNum(this.startField) !== null &&
127            fieldNum(this.endField) !== null
128        );
129    }
130
131    refreshNextField(): void {
132        this.nextField = String(this.nextRallyNumber());
133    }
134
135    lastRow(): RallyRow | null {
136        return this.rows.length ? this.rows[this.rows.length - 1] : null;
137    }
138
139    private editedRally(): RallyRow | null {
140        return this.mode === "edit" && this.editIndex !== null ? (this.rows[this.editIndex]
141    ?? null) : null;
142    }
143
144    // ---- button labels (port of refresh_buttons) -> translatable Msg ----
145
146    saveLabel(): Msg {
147        const r = this.editedRally();
148        if (r) return { key: "btn.saveChangesN", params: { n: r.n } };
149        const s = fieldNum(this.startField);
150        const e = fieldNum(this.endField);
151        if (s !== null && e !== null) return { key: "btn.saveRallyN", params: { n:
152    this.plannedNextNumber() } };
153        return { key: "btn.saveRally" };
154    }
155
156    markStartLabel(): Msg {
157        const r = this.editedRally();
158        return r ? { key: "btn.reMarkStart", params: { n: r.n } } : { key: "btn.markStart"
159    };

```

```

157     }
158
159     markEndLabel(): Msg {
160         const r = this.editedRally();
161         return r ? { key: "btn.reMarkEnd", params: { n: r.n } } : { key: "btn.markEnd" };
162     }
163
164     undoLabel(): Msg {
165         if (this.mode === "edit") return { key: "btn.undoCancelEdit" };
166         const s = fieldNum(this.startField);
167         const e = fieldNum(this.endField);
168         if (s !== null || e !== null) return { key: "btn.undoClearMark" };
169         const last = this.lastRow();
170         if (last) return { key: "btn.undoN", params: { n: last.n } };
171         return { key: "btn.undo" };
172     }
173
174     // ---- form reset: reason + shots are NON-STICKY; sport & nextField untouched ----
175     resetForm(): void {
176         this.mode = "new";
177         this.editIndex = null;
178         this.startField = "";
179         this.endField = "";
180         this.shotsField = "";
181         this.reason = REASON_DEFAULT;
182     }
183
184     // ---- operations (port of the button callbacks) -> translatable OpResult ----
185
186     markStart(now: number | null): OpResult {
187         if (now == null) return { ok: false, key: "status.noMediaStart" };
188         if (this.isArmed()) return { ok: false, key: "status.unsavedGuard" };
189         this.startField = now.toFixed(3);
190         return { ok: true, key: "status.startSet", params: { t: now.toFixed(3) } };
191     }
192
193     markEnd(now: number | null): OpResult {
194         if (now == null) return { ok: false, key: "status.noMediaEnd" };
195         this.endField = now.toFixed(3);
196         return { ok: true, key: "status.endSet", params: { t: now.toFixed(3) } };
197     }
198
199     saveRally(): OpResult {
200         let s = fieldNum(this.startField);
201         let e = fieldNum(this.endField);
202         if (s == null) return { ok: false, key: "status.needStart" };
203         if (e == null) return { ok: false, key: "status.needEnd" };
204         if (e < s) {
205             const t = s;
206             s = e;
207             e = t;
208         } // tolerate reversed marks
209         if (e <= s) return { ok: false, key: "status.zeroLength" };
210
211         const reason = this.reason || REASON_DEFAULT;
212         const sport = this.sport || SPORTS[0];
213         const shotsNum = this.getShots();
214         const shots = shotsNum !== null ? String(shotsNum) : null;
215
216         if (this.mode === "edit" && this.editIndex !== null && this.rows[this.editIndex]) {
217             const r = this.rows[this.editIndex];
218             const backup = { ...r };
219             r.s = s;
220             r.e = e;
221             r.reason = reason;

```

```

222         r.sport = sport;
223         r.shots = shots;
224         const res = this.persist(this.rows);
225         if (!res.ok) {
226             Object.assign(r, backup);
227             return { ok: false, key: "status.writeFailed", params: { err: String(res.err) }
};
228         }
229         const params = { n: r.n, s: s.toFixed(3), e: e.toFixed(3), reason, sport };
230         this.resetForm();
231         return { ok: true, key: "status.updated", params };
232     }
233
234     const n = this.plannedNextNumber();
235     if (this.indexOfRally(n) !== -1) return { ok: false, key: "status duplicate",
params: { n } };
236     const row: RallyRow = { n, s, e, reason, sport, shots, extra: null };
237     this.rows.push(row);
238     const res = this.persist(this.rows);
239     if (!res.ok) {
240         this.rows.pop();
241         return { ok: false, key: "status.writeFailed", params: { err: String(res.err) } };
242     }
243     this.nextField = String(this.nextFreeFrom(n + 1));
244     const params = { n, s: s.toFixed(3), e: e.toFixed(3), reason, sport };
245     this.resetForm();
246     return { ok: true, key: "status.saved", params };
247 }
248
249 editSelected(n: number | null): OpResult {
250     if (this.isArmed()) return { ok: false, key: "status.editGuard" };
251     if (n == null) return { ok: false, key: "status.needSelectEdit" };
252     const idx = this.indexOfRally(n);
253     if (idx === -1) return { ok: false, key: "status.notFound", params: { n } };
254     const r = this.rows[idx];
255     this.mode = "edit";
256     this.editIndex = idx;
257     this.startField = r.s.toFixed(3);
258     this.endField = r.e.toFixed(3);
259     this.shotsField = r.shots != null ? r.shots : "";
260     this.reason = r.reason || REASON_DEFAULT;
261     this.sport = r.sport || this.sport;
262     return { ok: true, key: "status.editing", params: { n: r.n } };
263 }
264
265 deleteSelected(n: number | null): OpResult {
266     if (n == null) return { ok: false, key: "status.needSelectDelete" };
267     const idx = this.indexOfRally(n);
268     if (idx === -1) return { ok: false, key: "status.notFound", params: { n } };
269     const removed = this.rows.splice(idx, 1)[0];
270     const res = this.persist(this.rows);
271     if (!res.ok) {
272         this.rows.splice(idx, 0, removed);
273         return { ok: false, key: "status.writeFailed", params: { err: String(res.err) } };
274     }
275     this.resetForm();
276     this.refreshNextField();
277     return { ok: true, key: "status.deleted", params: { n, count: this.rows.length } };
278 }
279
280 // 3-way: cancel edit / clear in-progress mark / drop the last committed row.
281 undoLast(): OpResult {
282     if (this.mode === "edit") {
283         this.resetForm();
284         return { ok: true, key: "status.editCancelled" };

```



```

285     }
286     const s = fieldNum(this.startField);
287     const e = fieldNum(this.endField);
288     if (s !== null || e !== null) {
289         this.resetForm();
290         return { ok: true, key: "status.clearedMark" };
291     }
292     if (this.rows.length === 0) return { ok: false, key: "status.nothingUndo" };
293     const last = this.rows.pop();
294     const res = this.persist(this.rows);
295     if (!res.ok) {
296         this.rows.push(last);
297         return { ok: false, key: "status.writeFailed", params: { err: String(res.err) } };
298     }
299     this.refreshNextField();
300     return { ok: true, key: "status.removedLast", params: { n: last.n, count:
this.rows.length } };
301     }
302
303     toCSV(): string {
304         return serializeRows(this.rows);
305     }
306 }
307

```

13. user (2026-07-02T02:48:05.910Z)

```

1      // Live store + per-video identity. chrome.storage.local is the source of truth (it is
2      // extension-scoped ? unlike a content script's IndexedDB, which would live in the
3      // VISITED site's origin ? and is plenty for per-video rally data). Resume numbering is
4      // derived from the loaded rows by the Annotator, exactly like VLC's
next_rally_number().
5
6      import { browser } from "wxt/browser";
7      import type { RallyRow } from "../state/csv";
8
9      const PREFIX = "rally:";
10
11      export interface VideoIdentity {
12          key: string; // storage discriminator (stable per video)
13          title: string; // human label, used as the CSV filename base
14      }
15
16      // Structural subset of Location so this is unit-testable with a plain URL.
17      export interface LocationLike {
18          hostname: string;
19          search: string;
20          pathname: string;
21          origin: string;
22      }
23
24      // Derive a stable key for the current page's video. YouTube watch/shorts/youtu.be get
25      // the video id; everything else uses origin+pathname (volatile query/hash dropped).
26      // location/title are injectable so this is testable without a DOM.
27      export function deriveIdentity(loc: LocationLike = location, docTitle?: string):
VideoIdentity {
28          const host = loc.hostname.replace(/^www\./, "");
29          const fallbackTitle = typeof document !== "undefined" ? document.title : "";
30          const title = ((docTitle ?? fallbackTitle) || host || "rallies").trim();
31
32          if (/^(^|\.)youtube\.com$/.test(host)) {
33              const v = new URLSearchParams(loc.search).get("v");
34              const shorts = loc.pathname.match(/^\/shorts\/([^\s]+)/);
35              const id = v ?? (shorts ? shorts[1] : null);
36              if (id) return { key: `yt:${id}`, title };

```

```

37     }
38     if (host === "youtu.be") {
39         const id = loc.pathname.slice(1).split("/")[0];
40         if (id) return { key: `yt:${id}`, title };
41     }
42     return { key: `url:${loc.origin}${loc.pathname}`, title };
43 }
44
45 export async function loadRows(key: string): Promise<RallyRow[]> {
46     const r = await browser.storage.local.get(PREFIX + key);
47     const v = (r as Record<string, unknown>)[PREFIX + key];
48     return Array.isArray(v) ? (v as RallyRow[]) : [];
49 }
50
51 export async function saveRows(key: string, rows: RallyRow[]): Promise<void> {
52     await browser.storage.local.set({ [PREFIX + key]: rows });
53 }
54
55 // Sanitize a title into a safe "<name>.rallies.csv" filename (mirrors the VLC output
56 // naming of <video-stem>.rallies.csv, as close as a sandboxed download allows).
57 export function csvFilename(title: string): string {
58     const base = title
59         .replace(/[\\/:*?"<>|]+/g, "_")
60         .replace(/\s+/g, " ")
61         .trim()
62         .slice(0, 80);
63     return base ? `${base}.rallies.csv` : "rallies.csv";
64 }
65

```

14. user (2026-07-02T02:48:06.849Z)

```

1     import { browser } from "wxt/browser";
2     import { Annotator } from "../src/state/annotator";
3     import { DirectVideoHandler } from "../src/video/directVideo";
4     import { deriveIdentity, loadRows, saveRows, csvFilename } from "../src/persist/store";
5     import { mountPanel } from "../src/ui/panel";
6     import type { RuntimeMessage } from "../src/messages";
7
8     // Top-frame content script: derives the per-video identity, loads any saved rallies,
9     // wires the pure Annotator to the page's video + storage + CSV download, and mounts the
10    // panel (hidden until the toolbar icon is clicked). Cross-origin EMBEDDED players
11    // (youtube.com/embed inside a third-party page) are a later, separate handler.
12    export default defineContentScript({
13        matches: ["<all_urls>"],
14        runAt: "document_idle",
15        async main() {
16            if (window.top !== window) return; // v1: top frame only
17
18            const identity = deriveIdentity();
19            const video = new DirectVideoHandler();
20            const rows = await loadRows(identity.key);
21
22            const annotator = new Annotator({
23                rows,
24                // chrome.storage.local is the source of truth (extension-scoped). Async mirror;
25                // the in-memory annotator.rows is authoritative for the live session.
26                persist: (rs) => {
27                    void saveRows(identity.key, rs);
28                    return { ok: true };
29                },
30            });
31
32            const panel = mountPanel({
33                annotator,

```

```

34         video,
35         identity,
36         download: (csv) =>
37             void browser.runtime.sendMessage({
38                 type: "rally-download",
39                 filename: csvFilename(identity.title),
40                 csv,
41             } as RuntimeMessage),
42     });
43
44     panel.toggle(); // start hidden; the toolbar icon reveals it on pages you want to
annotate
45
46     browser.runtime.onMessage.addListener((msg: RuntimeMessage) => {
47         if (msg && msg.type === "toggle-panel") panel.toggle();
48     });
49 },
50 });
51

```

15. user (2026-07-02T02:48:07.811Z)

```

web/src/i18n/locales/da/common.json:2:   "_meta": {
web/src/i18n/locales/en/common.json:2:   "_meta": { "locale": "en", "status": "source" },
web/src/i18n/locales/es/common.json:2:   "_meta": {
web/src/i18n/locales/hi/common.json:2:   "_meta": {
web/src/i18n/locales/id/common.json:2:   "_meta": {
web/src/i18n/locales/kn/common.json:2:   "_meta": {
web/src/i18n/locales/te/common.json:2:   "_meta": {

```

16. user (2026-07-02T02:48:14.575Z)

```

4:   "status": "machine-draft",
52: "status.ready": "?????? ??? ?????, ??? ??????? ????? ?????? ?? ??? ??? ?? ?????
?????",
53: "status.noMediaStart": "??? ?????? ????? ?? ??? ? ????? ?????? ????? ?? ?????",
6

```

17. user (2026-07-02T02:48:16.442Z)

C:/Users/avidu/Projects/khelsutra-guru/Annotation Setup/Archive:

```

total 8
drwxr-xr-x 1 avidu 197609 0 Jun 23 09:53 .
drwxr-xr-x 1 avidu 197609 0 Jun 23 09:53 ..
drwxr-xr-x 1 avidu 197609 0 Jun 23 09:48 Roora Pilot Review - 6 June 2026

```

C:/Users/avidu/Projects/khelsutra-guru/Annotation Setup/Golden Labelled:

```

total 33
drwxr-xr-x 1 avidu 197609 0 Jun 22 15:12 .
drwxr-xr-x 1 avidu 197609 0 Jun 23 09:53 ..
-rw-r--r-- 1 avidu 197609 4106 Jun 22 15:12 Adarsh and Avi.rallies.csv
-rw-r--r-- 1 avidu 197609 1413 Jun 22 15:12 KushagraYashAviFirstVideo.rallies.csv
-rw-r--r-- 1 avidu 197609 2140 Jun 22 15:05 LargeTestVideo.rallies.csv
-rw-r--r-- 1 avidu 197609 1325 Jun 22 15:05 MahadevpuraSingles.rallies.csv
-rw-r--r-- 1 avidu 197609 287 Jun 22 15:12 TestLargeVideo.rallies.csv
-rw-r--r-- 1 avidu 197609 2049 Jun 22 15:12 gBaaddy20May2026.rallies.csv

```

C:/Users/avidu/Projects/khelsutra-guru/Annotation Setup/Trajectories:

```

total 17960
drwxr-xr-x 1 avidu 197609 0 Jun 18 12:26 .
drwxr-xr-x 1 avidu 197609 0 Jun 23 09:53 ..
-rw-r--r-- 1 avidu 197609 1549493 Jun 17 15:02 Badminton_BXH_2_june15_traj.csv
-rw-r--r-- 1 avidu 197609 838358 Jun 18 11:28 Boxhill_Doubles_june18_traj.csv
-rw-r--r-- 1 avidu 197609 1543847 Jun 16 19:04 GX010128_june12_traj.csv

```

```

-rw-r--r-- 1 avidu 197609 797173 Jun 17 18:13 GX010137_june17_traj.csv
-rw-r--r-- 1 avidu 197609 776018 Jun 17 17:23 GX010141_june17_traj.csv
-rw-r--r-- 1 avidu 197609 608149 Jun 18 11:53 GX020094_june18_traj.csv
-rw-r--r-- 1 avidu 197609 701986 Jun 17 14:15 GX030094_june15_traj.csv
-rw-r--r-- 1 avidu 197609 2476375 Jun 8 18:24 adarsh_avi_full_traj.csv
-rw-r--r-- 1 avidu 197609 1114525 Jun 8 14:24 adarsh_avi_traj.csv
-rw-r--r-- 1 avidu 197609 1101363 Jun 6 20:53 adarsh_traj.csv
-rw-r--r-- 1 avidu 197609 1702622 Jun 8 19:11 gbaaddy_traj.csv
-rw-r--r-- 1 avidu 197609 1159102 Jun 8 18:42 kushagra_traj.csv
-rw-r--r-- 1 avidu 197609 693876 Jun 8 21:02 largetestvideo_traj.csv
-rw-r--r-- 1 avidu 197609 459324 Jun 16 00:58 mahadevpura1_june12_traj.csv
-rw-r--r-- 1 avidu 197609 1454557 Jun 16 02:26 mahadevpura2_june12_traj.csv
-rw-r--r-- 1 avidu 197609 418680 Jun 9 10:40 mahadevpura_traj.csv
-rw-r--r-- 1 avidu 197609 145287 Jun 9 18:10 pickleball_traj.csv
drwxr-xr-x 1 avidu 197609 0 Jun 10 12:56 roora
-rw-r--r-- 1 avidu 197609 600480 Jun 6 18:01 sample_long_traj.csv
-rw-r--r-- 1 avidu 197609 96822 Jun 9 18:13 tabletennis_traj.csv
-rw-r--r-- 1 avidu 197609 96359 Jun 6 22:20 testlarge_traj.csv
rally_number,start_mmss,end_mmss,start_time,end_time,shots_count,ending_reason,last_shot_outcome
,notes,status
1,0:03.003,0:21.021,3.003,21.021,15,,net,"Roora put ending_reason=other; net is an outcome, set
forced/unforced",needs_review
2,0:29.000,0:32.533,29.000,32.533,2,,,,"start corrected from 0:02 (stray early box); end 0:32 ok
-> ~3s rally. Also removes the 1<->2 overlap",CORRECTED-start
3,0:44.545,0:51.051,44.545,51.051,5,unforced_error,,,needs_review
4,1:02.062,1:07.067,62.062,67.067,4,unforced_error,,,needs_review
rally,start_mmss,end_mmss,dur_s,shots,ending_reason,ending_consistent,notes,auto_flag,VERIFIED(Y
/N),true_start_mmss,true_end_mmss,true_ending,reviewer_comment
1,0:03.003,0:21.021,18.0,15,other,yes,net,,,,,,,,
2,0:02.502,0:32.532,30.0,2,forced_error,yes,,OVERLAP;INFLATED?,,,,,
3,0:44.544,0:51.051,6.5,5,unforced_error,yes,,,,,,,,,
4,1:02.062,1:07.067,5.0,4,unforced_error,yes,,,,,,,,,

```

18. user (2026-07-02T02:48:29.781Z)

```

1      {
2        "name": "rally-annotator-web",
3        "version": "0.1.0",
4        "private": true,
5        "type": "module",
6        "description": "Browser-extension port of the Rally Annotator VLC tool (Chrome-first,
Manifest V3, cross-browser-ready).",
7        "scripts": {
8          "dev": "wxt",
9          "build": "wxt build",
10         "build:firefox": "wxt build -b firefox",
11         "zip": "wxt zip",
12         "prepare:wxt": "wxt prepare",
13         "test": "vitest run",
14         "test:watch": "vitest",
15         "coverage": "vitest run --coverage",
16         "e2e": "playwright test",
17         "e2e:install": "playwright install chromium",
18         "typecheck": "tsc --noEmit"
19       },
20       "devDependencies": {
21         "@playwright/test": "^1.61.0",
22         "@vitest/coverage-v8": "^3.2.6",
23         "jsdom": "^29.1.1",
24         "typescript": "^5.7.3",
25         "vitest": "^3.2.4",
26         "wxt": "^0.20.26"
27       }
28     }
29

```

19. user (2026-07-02T02:48:30.025Z)

```
badminton-highlight-indexer/output/Badminton_BXH_2.rallies.csv
badminton-highlight-indexer/output/Boxhill_Doubles.rallies.csv
badminton-highlight-indexer/output/GX010128.rallies.csv
badminton-highlight-indexer/output/GX010137.rallies.csv
badminton-highlight-indexer/output/GX010141.rallies.csv
badminton-highlight-indexer/output/GX020094.rallies.csv
badminton-highlight-indexer/output/GX030094.rallies.csv
badminton-highlight-indexer/output/mahadevpura_1.rallies.csv
badminton-highlight-indexer/output/mahadevpura_2.rallies.csv
badminton-highlight-indexer/scratch/stage/Badminton BHX 2_proxy.rallies.csv
badminton-highlight-indexer/scratch/stage/Boxhill Doubles_proxy.rallies.csv
badminton-highlight-indexer/scratch/stage/GX010128.rallies.csv
badminton-highlight-indexer/scratch/stage/GX020094_proxy.rallies.csv
badminton-highlight-indexer/scratch/stage/GX030094_proxy.rallies.csv
badminton-highlight-indexer/scratch/stage/mahadevpura-1.rallies.csv
badminton-highlight-indexer/scratch/stage/mahadevpura-2_proxy.rallies.csv
---
badminton-highlight-indexer/output/wasb_tuned_segments_vs_golden.csv
```

20. user (2026-07-02T02:48:38.800Z)

```
output/Badminton_BXH_2.rallies.csv
IGNORED
backend/eval/fusion_golden.py
backend/eval/golden_fixtures.py
backend/eval/golden_real_fixtures.py
docs/GOLDEN_REGRESSION_FIXTURES.md
docs/archives/roora-vendor/GOLDEN_SET_VENDORING.md
docs/data_pipeline/GOLDEN_DATA_SHARING.md
docs/data_pipeline/GOLDEN_SET_IMPLEMENTATION.md
docs/data_pipeline/GOLDEN_SET_PHASE1_VIDEOS.md
docs/data_pipeline/GOLDEN_VIDEOS.md
tests/test_fusion_golden_telemetry.py
tests/test_golden_fixtures.py
tests/test_golden_fixtures_split.py
tests/test_golden_real_fixtures.py
```

21. user (2026-07-02T02:48:40.848Z)

```
rally_number,start_time,end_time,ending_reason,sport
1,37.434,72.194,winner,badminton
2,79.936,112.954,winner,badminton
Annotation Videos
Feedback
Illustrated Annotation Guide ? Rally Labeling (Supplement).pdf
Roora Annotation Brief ? Rally Golden Set (Format Spec, Scope, QA).pdf
Roora Pilot Review ? Badminton (2026-06-06).md
Roora Pilot Review ? Pickleball & Table Tennis (2026-06-10).md
badminton
pickleball
tabletennis
```

22. user (2026-07-02T02:48:51.767Z)

Golden / GPU-Extractor Data Sharing ? GPU box ? dev sessions

```
**Status:** convention + plan (2026-06-15). Tracked in **#175**.
**Purpose:** a single cloud-synced location for the GPU-extracted golden artifacts (trajectory
CSVs, `*_golden_features.json`, manifests, baseline snapshots) so the **GPU box** and **CPU
dev/agent sessions** share the same files ? without committing large data to git.
```

Why this exists

The rally-quality measurement is split across two machines by capability:

- `backend/eval/fusion_golden.py` (shuttle+person feature extraction) needs the `videos`, the `trajectory CSVs`, the `.rallies.csv` golden labels, and a `GPU/WSL` stack ? only the GPU box can run it.
- `backend/eval/fusion_compare.py` / `backend/eval/ablation.py` (the LOVO litmus: ?F1, bootstrap CI, paired sign test) are `pure-CPU re-scoring` that read only the small `_golden_features.json` ? any session (incl. a CPU dev container) can run them.

Today those artifacts sit in gitignored `output/` (per-machine) and scattered local folders (`...\Annotation Setup\Collect\Trajectories\`, `...\Golden Labelled\`). Nothing syncs them between machines, so a CPU session cannot see the GPU box's extraction. This wires that gap with OneDrive.

Canonical shared folder

```
%USERPROFILE%\OneDrive\rally-annotator\golden-data\ ? C:\Users\avidu\OneDrive\rally-annotator\golden-data\
```

OneDrive is present + signed-in on both the GPU box and the dev machine under the same account, so it auto-syncs. Layout (date/size-tagged so a re-extraction is a new folder, never an overwrite):

```
...
```

```
golden-data/
  README.md
  features/<corpus-tag>/      *_golden_features.json      (small ? the key shared artifact)
  trajectories/<corpus-tag>/  *_predict.csv / *_wasb.csv (medium ? reusable across runs)
  manifests/                 *_lovo_manifest.json
  baselines/                 leaderboard / regression snapshots (or keep tracked in
eval_baselines/)
...
```

`<corpus-tag>` = e.g. `2026-06-15-n6` (date + corpus size). `Do NOT` put raw videos here (tens of GB) ? keep those local to the GPU box.

The GPU ? CPU workflow

1. `GPU box` (videos + CSVs + labels present):

```
...
```

```
python -m backend.eval.fusion_golden --manifest <manifest> \
  --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-data\features\2026-06-15-n6"
...
```

(or extract to local `output/` then copy the `_golden_features.json` into the shared folder).

2. OneDrive syncs the JSONs to the dev machine automatically (small files, seconds).
3. `CPU dev/agent session` points the analysis at the shared folder:

```
...
```

```
python -m backend.eval.fusion_compare --features-dir "...golden-data\features\2026-06-15-n6"
--record
python -m backend.eval.ablation          --features-dir "...golden-data\features\2026-06-15-n6"
--granularity group
...
```

? real person-feature ?F1 / CI / sign test. Copy the JSONs into the repo's `output/` to also light up the `skipif` golden-litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`, `tests/test_fusion_compare.py`).

Optional code ergonomics (in #175, not required)

- A `RALLY_GOLDEN_DIR` env var (default `output/`) honored by `fusion_golden` / `fusion_compare` / `ablation` so neither machine hardcodes the path. (The CLI `--out-dir` / `--features-dir` flags already make this a convention; the env var is just ergonomics.)
- A tiny `scripts/sync_golden.py` to copy `output/_golden_features.json` ? the shared folder.

Guardrails

- The shared folder lives `outside the repo` ? never committed. Keep `output/` gitignored and

do ****not**** symlink the shared folder into a tracked path.
- The features derive from the ****Proprietary**** corpus ? keep the OneDrive folder ****private**** (no public share links).

23. user (2026-07-02T02:48:53.752Z)

```
backend/eval/export_wasb_segments.py
docs/archives/quality-iterations/2026-06-human-gt-tuning/README.md
docs/CODE_MAP.md
docs/DATA_IN_GCS.md
docs/data_pipeline/GOLDEN_DATA_SHARING.md
docs/NEXT_STEPS.md
scratch/migrate_golden_to_gcs.py
scratch/p3_smoke.py
60
"""Golden Regression Fixtures (P1 + M2) ? de-attribution / minimization + the refusal gate.
```

This module holds the ****real-fixture subsystem**** split out of ```golden_fixtures``` (which keeps the shared synthetic/replay primitives). It turns an OWNED, minors-free real trajectory + golden labels into a committable, de-attributed Tier-0 fixture, reusing M1's ```replay_fixture``` for the snapshot.
(docs/GOLDEN_REGRESSION_FIXTURES.md §4c anti-attribution, §5 threat model, §7 row P1, §6 honest limits.)

The shared primitives (constants, float/JSON helpers, ```replay_fixture```, the CSV writers, the manifest appender) live in ```golden_fixtures``` and are imported FROM it here ? there is a single source of truth, no duplicated logic/thresholds. ```golden_fixtures``` re-exports the public names defined here (```freeze_real_fixture```, ```RefusalError```, ```_scan_forbidden```) at the bottom of that module so every historical ```golden_fixtures.<name>``` import path stays valid.

§6 honest-limits caveat: a green suite proves only that the deterministic post-trajectory transforms are unchanged for the frozen cases ? characterization is behaviour-locking, NOT correctness.
"""

```
from __future__ import annotations

import os
import shutil
import tempfile
from pathlib import Path
from typing import Any, Dict, List, Optional, Sequence, Tuple
```

--- shared primitives, imported FROM golden_fixtures (single source of truth) -----

```
from backend.eval.golden_fixtures import (
    _LABEL_SLUG_RE,
    _MD5_RE,
    CURRENT_SCHEMA,
    FIXTURES_DIR,
    FORBIDDEN_KEYS,
    PINNED_MAX_WINDOW,
    PINNED_MERGE_GAP,
    PINNED_MIN_WINDOW,
    PINNED_VT,
    Interval,
    _append_manifest_entry,
    _round,
```

24. user (2026-07-02T02:49:02.527Z)

```
49:| `labels/` | golden rally labels (`<clip>.rallies.csv`) | **input** ground truth to `worker
```

```

train` + all eval |
68:> **Phase-1 landed 2026-06-22:** added `trajectories/` (15), 15 canonical
`labels/<date>_<name>.rallies.csv`
121:| 20 WASB trajectory CSVs | `C:\?\Annotation Setup\Collect\Trajectories\` | **17.5 MB** | ?
none | eval/litmus (cached). *(No `cached` train source exists ? train synths or detector-
extracts.)* |
123:| original-6 labels | `F:\[Khelsutra]\Training\Golden Labelled\*.rallies.csv` | ~12 KB | ?
| train + eval |
124:| 7 new-clip labels | repo `output/*.rallies.csv` (gitignored) **** F: archive (backed up
2026-06-22) | ~12 KB | ? partial/inconsistent | train + eval |
143:# labels (after the §7.1 naming decision) -> labels/<canonical-name>.rallies.csv
182:1. **`labels/` naming ? LOCKED (owner, 2026-06-22):** `labels/<YYYY-MM-
DD>_<name>.rallies.csv`, where
199:> ? **Cleanup DONE 2026-06-22:** the 7 pre-existing inconsistent label objects
(`*_proxy.rallies.csv`,
214: join key as `fname.replace(".rallies.csv","")` ? `2026-06-22_GX020094`, then looked up
218: `.rallies.csv/`.csv` suffix AND a leading `^\\d{4}-\\d{2}-\\d{2}_` prefix, so a dated label
and a legacy

```

25. user (2026-07-02T02:49:02.569Z)

```

1      # Rally Annotator ? browser extension (web port)
2
3      A Manifest V3 browser extension that brings the [VLC rally
annotator](../vlc/rally_annotator.lua)
4      to **web video**: open a page with a playing HTML5 `<video>`, mark each rally's
START/END +
5      ending reason, and get the **same CSV**
(`rally_number,start_time,end_time,ending_reason,sport,shots_count`)
6      that feeds the Khelsutra rally-segmentation pipeline.
7
8      > Part of **[Khelsutra](https://khelsutra.guru)** ? _?Every rally, indexed. The dead
time, gone.?_ This is
9      > the web front-end of Khelsutra's open rally-labeling tool. **Docs:**
[DESIGN.md](DESIGN.md) (architecture +
10     > locked decisions) · [TESTING.md](TESTING.md) (test strategy + per-browser proof) ·
[../CONTRIBUTING.md](../CONTRIBUTING.md).
11
12     **Chrome-first, loaded unpacked.** Built as one cross-browser-ready codebase (WXT);
Firefox + Safari
13     + store publishing are the north star (low-rework follow-ups), not yet wired.
14
15     ## Status (v0.1.0)
16
17     Works now:
18     - Full v1.7.1 dialog parity: two-step Mark?Save, non-sticky reason (default `unknown`),
sticky
19     sport, editable Start/End, `Next rally #` continuity + auto-advance, **Number of
shots**, 3-way
20     Undo, edit-mode **Re-mark (#N)** relabel, **unsaved-rally guard**, recent-rallies list
(all,
21     oldest-first), Back 5s / Play-Pause / Fwd 5s.
22     - **Localized into 7 languages** (English, Hindi, Kannada, Telugu, Spanish, Danish,
Indonesian) with a
23     persisted in-panel language selector ? machine-draft translations pending native
review (see
24     [../docs/LOCALIZATION.md](../docs/LOCALIZATION.md)).
25     - Controls any **direct HTML5 `<video>`** in the top frame, including videos in
(open/closed)
26     shadow DOM and **youtube.com/watch** (its player is in the main page).
27     - Rallies autosave to extension storage keyed per video ? reloading the page resumes
numbering.
28     - The full CSV downloads on each Save (and via the **Download CSV** button).
29
30     Not yet (deliberate follow-ups):

```



```

31 - **YouTube/Vimeo embeds on third-party pages** (cross-origin `<iframe>`) ? needs a per-
provider
32 handler injected into the provider frame.
33 - Firefox/Safari build legs and store publishing.
34 - `showSaveFilePicker` single-file "rewrite one CSV" mode (Chrome/Edge only).
35
36 ## Load it in Chrome
37
38 ```
39 cd web
40 npm install
41 npm run build      # outputs web/.output/chrome-mv3
42 ```
43
44 Then: `chrome://extensions` ? enable **Developer mode** ? **Load unpacked** ? select
45 `web/.output/chrome-mv3`. Open a page with a video and **click the extension's toolbar
icon** to
46 show/hide the panel. Drag it by its header; it stays visible over fullscreen video
(desktop).
47
48 `npm run dev` runs WXT in watch mode (auto-reload) if you prefer.
49
50 ## Develop / test
51
52 Proof-and-validation driven ? see [TESTING.md](TESTING.md) for the full strategy and the
53 per-browser proof matrix.
54
55 ```
56 npm test          # Vitest: CSV round-trip (byte-identical to the VLC tool) + state-
machine + DOM
57 npm run coverage  # unit + enforced coverage thresholds (~98% stmts/lines/fun
58 npm run typecheck
59 npm run build
60 npm run e2e       # real-Chromium E2E that loads the built extension and records a
screencast
61 ```
62
63 CI (.github/workflows/web-ci.yml) runs all of the above on every change under `web/
and
64 uploads the E2E screencast + coverage as artifacts.
65
66 ## Architecture
67
68 - `src/state/csv.ts` ? CSV serialize/parse, **byte-compatible** with the Lua
save_all/load_rows.
69 - `src/state/annotator.ts` ? pure state machine ported ~1:1 from the Lua callbacks
(numbering, undo,
70 guards, labels); unit-tested, no browser needed.
71 - `src/video/directVideo.ts` ? finds/controls the active `<video>` (seconds, no  $\mu$ s
conversion).
72 - `src/persist/store.ts` ? `chrome.storage.local` live store + per-video identity + CSV
filename.
73 (Extension-scoped on purpose ? a content script's IndexedDB would live in the visited
site's
74 origin. IndexedDB-in-service-worker is the path if storage ever needs to scale.)
75 - `src/ui/panel.ts` ? open-shadow-DOM panel reproducing the VLC widget set (open = same
CSS
76 isolation as closed, but reachable by automation/devtools).
77 - `entrypoints/content.ts` ? top-frame bootstrap; `entrypoints/background.ts` ? CSV
download + icon toggle.
78
79 ## Output
80
81 The CSV is downloaded to your **Downloads** folder as `<page-title>.rallies.csv` (a
sandboxed

```

```
82     extension can't write next to a web video). It is identical in schema to the VLC tool's
output ? see
83     [../docs/CSV_FORMAT.md](../docs/CSV_FORMAT.md).
84
```

26. user (2026-07-02T02:49:28.296Z)

```
origin https://github.com/avidullu/rally-annotator.git (fetch)
origin https://github.com/avidullu/rally-annotator.git (push)
2026-06-22 11:37:07 +0530 (HEAD -> main, origin/main)
```

27. user (2026-07-02T02:49:28.493Z)

```
1     import { browser } from "wxt/browser";
2     import type { RuntimeMessage } from "../src/messages";
3
4     // Background service worker: performs the CSV download (chrome.downloads is unavailable
5     // to content scripts) and routes the toolbar-icon click into a panel toggle.
6     export default defineBackground(() => {
7         browser.runtime.onMessage.addListener((msg: RuntimeMessage) => {
8             if (msg && msg.type === "rally-download") {
9                 const url = "data:text/csv;charset=utf-8," + encodeURIComponent(msg.csv);
10                 void browser.downloads
11                     .download({ url, filename: msg.filename, saveAs: false, conflictAction:
"overwrite" })
12                     .catch((e) => console.error("[rally] download failed:", e));
13             }
14         });
15
16         // No default_popup is set, so the icon click fires onClicked -> toggle the panel.
17         browser.action.onClicked.addListener((tab) => {
18             if (tab.id !== null) {
19                 void browser.tabs
20                     .sendMessage(tab.id, { type: "toggle-panel" } as RuntimeMessage)
21                     .catch(() => {
22                         /* tab has no content script (e.g. chrome:// page) */
23                     });
24             }
25         });
26     });
27
```

28. user (2026-07-02T02:49:29.101Z)

```
1     // Finds and controls the active HTML5 <video> on the page ? including videos nested
2     // in (open or closed) shadow DOM ? and exposes the playback primitives the annotator
3     // needs: the web analogue of VLC's now_seconds() / seek_by() / play_pause().
4     //
5     // Key simplification vs VLC: HTML5 video.currentTime is already in SECONDS, so there
6     // is no microsecond conversion. This handler covers same-origin/top-frame videos
7     // (including youtube.com/watch, whose player is in the main page). Cross-origin
8     // EMBEDDED players (e.g. a youtube.com/embed iframe on a third-party site) are a
9     // separate, later handler ? they live in a frame this top-frame script can't reach.
10
11     // chrome.dom.openOrClosedShadowRoot lets a content script pierce CLOSED shadow roots
12     // (Chrome/Firefox); fall back to the public .shadowRoot otherwise.
13     function openOrClosed(el: Element): ShadowRoot | null {
14         const anyChrome = (globalThis as unknown as { chrome?: any }).chrome;
15         try {
16             if (anyChrome?.dom?.openOrClosedShadowRoot) {
17                 return anyChrome.dom.openOrClosedShadowRoot(el) ?? null;
18             }
19         } catch {
20             /* ignore */
21         }
22     }
23
```

```

21     }
22     return (el as unknown as { shadowRoot?: ShadowRoot | null }).shadowRoot ?? null;
23 }
24
25 function collectVideos(root: ParentNode, acc: HTMLVideoElement[], depth: number): void {
26     if (depth > 8) return; // bound the shadow-DOM walk
27     root.querySelectorAll("video").forEach((v) => acc.push(v as HTMLVideoElement));
28     root.querySelectorAll("*").forEach((el) => {
29         const sr = openOrClosed(el as Element);
30         if (sr) collectVideos(sr, acc, depth + 1);
31     });
32 }
33
34 export class DirectVideoHandler {
35     private current: HTMLVideoElement | null = null;
36
37     findActive(): HTMLVideoElement | null {
38         const vids: HTMLVideoElement[] = [];
39         collectVideos(document, vids, 0);
40         if (vids.length === 0) {
41             this.current = null;
42             return null;
43         }
44         // Prefer a currently-playing video; tie-break by largest rendered area.
45         const score = (v: HTMLVideoElement) => {
46             const r = v.getBoundingClientRect();
47             const area = Math.max(0, r.width) * Math.max(0, r.height);
48             return (v.paused ? 0 : 1e12) + area;
49         };
50         vids.sort((a, b) => score(b) - score(a));
51         this.current = vids[0];
52         return this.current;
53     }
54
55     private active(): HTMLVideoElement | null {
56         if (this.current && this.current.isConnected) return this.current;
57         return this.findActive();
58     }
59
60     hasVideo(): boolean {
61         return this.active() != null;
62     }
63
64     now(): number | null {
65         const v = this.active();
66         return v ? v.currentTime : null;
67     }
68
69     duration(): number | null {
70         const v = this.active();
71         return v && Number.isFinite(v.duration) ? v.duration : null;
72     }
73
74     isPaused(): boolean | null {
75         const v = this.active();
76         return v ? v.paused : null;
77     }
78
79     seekBy(delta: number): boolean {
80         const v = this.active();
81         if (!v) return false;
82         let t = v.currentTime + delta;
83         if (t < 0) t = 0;
84         v.currentTime = t;
85         return true;

```

```

86     }
87
88     // Single toggle gated on actual paused state (mirrors the Lua play_pause()).
89     playPause(): "playing" | "paused" | null {
90         const v = this.active();
91         if (!v) return null;
92         if (v.paused) {
93             void v.play().catch(() => {});
94             return "playing";
95         }
96         v.pause();
97         return "paused";
98     }
99 }
100

```

29. assistant (2026-07-02T02:49:42.073Z)

The remote is `avidullu/rally-annotator` but README badges point to `Khelsutra/rally-annotator`. Let me check which org actually hosts the repo.

30. user (2026-07-02T02:49:45.024Z)

```

Khelsutra/rally-annotator | https://github.com/Khelsutra/rally-annotator
---
Khelsutra/rally-annotator | https://github.com/Khelsutra/rally-annotator

```

31. user (2026-07-02T02:49:53.667Z)

```

90:function descriptor()
743:-- live in function bodies, never at top level, so the descriptor() scan sandbox is fine.)
744:local function t(key, vars)
757:local function lang_index(code)
763:local function lang_config_path()
769:local function load_lang()
776:local function save_lang()
813:local function now_seconds()
822:local function fmt_clock(s)
831:local function esc(text)
840:local function current_media_path()
865:local function resolve_out_path()
885:local function load_rows()
922:local function save_all()
950:local function next_rally_number()
957:local function get_field_num(w)
968:local function get_shots()
977:local function is_armed()
983:local function planned_next_number()
990:local function refresh_next_field()
994:local function index_of_rally(n)
1001:local function next_free_from(start)
1009:local function selected_rally_number()
1024:local function rebuild_reason_default()
1032:local function rebuild_reason_selected(sel)
1044:local function rebuild_sport_selected(sel)
1059:local function refresh_list()
1074:local function refresh_buttons()
1118:local function set_status(msg)
1140:local function reset_form()
1155:local function sync_to_current_video()
1192:local function adopt_current_video_csv()
1206:local function seek_by(delta_s)
1220:function mark_start()
1233:function mark_end()

```

```

1241:function save_rally()
1288:function edit_selected()
1308:function delete_selected()
1322:function undo_last()
1345:function refresh_now()
1352:function play_pause()
1368:function seek_back() seek_by(-5) end
1369:function seek_fwd() seek_by(5) end
1371:function show_help()
1389:local function create_dialog()
1457:function change_language()
1473:function activate()
1489:function deactivate()
1494:function close()

```

32. user (2026-07-02T02:49:59.133Z)

```

813     local function now_seconds()
814         local input = vlc.object.input()
815         if not input then return nil end
816         local t_us = vlc.var.get(input, "time") -- MICROSECONDS in VLC 3.x
817         if not t_us then return nil end
818         return t_us / 1000000.0
819     end
820
821     -- mm:ss.mmm for display only.
822     local function fmt_clock(s)
823         if not s then return "--:--" end
824         if s < 0 then s = 0 end
825         local m = math.floor(s / 60)
826         local sec = s - m * 60
827         return string.format("%d:%06.3f", m, sec)
828     end
829
830     -- Minimal HTML-escape so paths/messages render literally in the rich-text label.
831     local function esc(text)
832         text = tostring(text or "")
833         text = text:gsub("&", "&amp;")
834         text = text:gsub("<", "&lt;")
835         text = text:gsub(">", "&gt;")
836         return text
837     end
838
839     -- Best-effort path to the currently playing media file (decoded from URI).
840     local function current_media_path()
841         local input = vlc.object.input()
842         if not input then return nil end
843         local item = vlc.input.item() -- VLC 3.x: vlc.input.item()
844         if not item then return nil end
845         local uri = item.uri() -- item.uri()
846         if not uri or uri == "" then return nil end
847         local p = uri
848         -- Strip the scheme. Windows file URIs look like file:///C:/dir/clip.mp4;
849         -- POSIX ones like file:///home/user/clip.mp4 -- keep the POSIX leading "/".
850         if p:match("^file:///a:") then
851             p = p:gsub("^file:///", "") -- file:///C:/... -> C:/...
852         else
853             p = p:gsub("^file://", "") -- file:///home/... -> /home/...
854         end
855         -- percent-decode (e.g. %20 -> space) BEFORE separator normalization
856         p = p:gsub("%%(%x%x)", function(h) return string.char(tonumber(h, 16)) end)
857         if p:match("^a:") then -- Windows drive path -> backslashes
858             p = p:gsub("/", "\\")
859         end
860         return p

```

```

861 end
862
863 -- Resolve a sensible default output path: next to the video if we can, else
864 -- the user's home (USERPROFILE on Windows, HOME elsewhere).
865 local function resolve_out_path()
866     local media = current_media_path()
867     if media then
868         local dir, stem = media:match("^([^\\\/])([\\\/]-)%.[^\\\/%.]*$")
869         if dir and stem and stem ~= "" then
870             return dir .. stem .. ".rallies.csv"
871         end
872         local d2 = media:match("^([^\\\/])")
873         if d2 then return d2 .. "rally_labels.csv" end
874     end
875     local home = os.getenv("USERPROFILE") or os.getenv("HOME") or "."
876     local sep = home:find("\\") and "\\" or "/"
877     if home:sub(-1) ~= sep then home = home .. sep end
878     return home .. "rally_labels.csv"
879 end
880
881 -- Load all rally rows from the CSV into `rows` (header + blank lines skipped).
882 -- Forgiving parser: any leading-integer line is a rally row. Column 6 is the
883 -- optional shots_count (blank/missing in older CSVs); any columns beyond it are
884 -- preserved verbatim so we never drop richer metadata on rewrite.
885 local function load_rows()
886     rows = {}
887     local f = io.open(out_path, "r")
888     if not f then return end
889     for line in f:lines() do
890         line = line:gsub("\r$", "")
891         if line:match("%S") then
892             local parts = {}
893             for field in (line .. ","):gmatch("([^\,]*)",) do parts[#parts + 1] = field end
894             local n = tonumber(parts[1])
895             if n and n == math.floor(n) then
896                 local shots = parts[6]
897                 if shots == "" then shots = nil end -- blank => not recorded
898                 local row = {
899                     n = math.floor(n),
900                     s = tonumber(parts[2]) or 0,
901                     e = tonumber(parts[3]) or 0,
902                     reason = (parts[4] ~= nil and parts[4] ~= "") and parts[4] or "other",
903                     sport = parts[5] or "",
904                     shots = shots,
905                 }
906                 if #parts > 6 then
907                     local ex = {}
908                     for i = 7, #parts do ex[#ex + 1] = parts[i] end
909                     row.extra = table.concat(ex, ",")
910                 end
911                 rows[#rows + 1] = row
912             end
913         end
914     end
915     f:close()
916 end
917
918 -- Rewrite the whole CSV from `rows` (header once). We write a temp file and swap
919 -- it into place so the live CSV is never lost if a step fails: rename is atomic on
920 -- POSIX (and succeeds for a brand-new file on Windows); when Windows can't rename
921 -- over an existing file we stash the original as .bak and roll back on error.
922 local function save_all()
923     local tmp = out_path .. ".tmp"
924     local f, err = io.open(tmp, "w")
925     if not f then return false, ("cannot open CSV for write: " .. tostring(err)) end

```

```

926     f:write(HEADER)
927     for _, r in ipairs(rows) do
928         local shots = (r.shots ~= nil) and tostring(r.shots) or ""
929         local line = string.format("%d,%.3f,%.3f,%s,%s,%s", r.n, r.s, r.e, r.reason,
r.sport, shots)
930         if r.extra and r.extra ~= "" then line = line .. "," .. r.extra end
931         f:write(line .. "\n")
932     end
933     f:close()
934     -- Atomic on POSIX; also succeeds on Windows when out_path doesn't exist yet.
935     if os.rename(tmp, out_path) then return true end
936     -- Windows overwrite path: keep a backup so the original is never lost.
937     local bak = out_path .. ".bak"
938     os.remove(bak)
939     local stashed = os.rename(out_path, bak)      -- move original aside (nil if none)
940     local ok, rerr = os.rename(tmp, out_path)
941     if not ok then
942         if stashed then os.rename(bak, out_path) end -- roll back
943         return false, ("cannot replace CSV: " .. tostring(rerr))
944     end
945     os.remove(bak)
946     return true
947 end
948
949 -- Next monotonic rally_number (max existing + 1); keeps numbering across reopen.
950 local function next_rally_number()
951     local maxn = 0
952     for _, r in ipairs(rows) do if r.n > maxn then maxn = r.n end end

```

33. user (2026-07-02T02:50:27.930Z)

```

1     import { defineConfig } from "wxt";
2
3     // One MV3 source tree; WXT emits per-browser bundles (Chrome now; Firefox/Safari are
4     // the north-star targets ? `wxt build -b firefox` etc.) and manages the background
5     // service-worker vs event-page split per browser. Version comes from package.json.
6     export default defineConfig({
7         manifest: {
8             name: "Rally Annotator",
9             description:
10                 "Mark rally start/end + a point-ending reason to a CSV while watching web video "
11                 +
12                 "(HTML5 <video>, incl. youtube.com/watch). Net-separated racquet sports.",
13             permissions: ["storage", "downloads"],
14             host_permissions: ["<all_urls>"],
15             action: { default_title: "Toggle Rally Annotator panel" },
16         },
17     });

```

34. user (2026-07-02T02:50:30.464Z)

GOLDEN_SET_VENDORING.md
README.md
ROORA_LABELING_GUIDELINES.md

C:/Users/avidu/Projects/khelsutra-guru/Annotation Setup/Archive/Roora Pilot Review - 6 June
2026/Feedback:
Rally8ShouldBeThis.png
Roora Pilot Review - Badminton (2026-06-06).pdf
Roora Pilot Review - Pickleball and Table Tennis (2026-06-10).pdf

C:/Users/avidu/Projects/khelsutra-guru/Annotation Setup/Archive/Roora Pilot Review - 6 June
2026/badminton:
badminton

badminton (1)
badminton (2)

35. user (2026-07-02T02:50:44.423Z)

20:The cloud engine itself was built by the (archived) DECOUPLED_COMPUTE project
21:(`docs/archives/past\_projects/DECOUPLED\_COMPUTE/`); this doc does **not** redesign it ? it records
124:| 7 new-clip labels | repo `output/\*.rallies.csv` (gitignored) **F**: archive (backed up 2026-06-22) | ~12 KB | ? partial/inconsistent | train + eval |
125:| source videos / proxies | **F**: archive + Google Drive golden folder | **100s** of GB | ? 3 only | infer + train (real WASB extraction) |
153:| **F**: archive (**full MP4s** ? proxy-first before upload) | `GX010137`, `GX010141`, + original-6 (`adarsh\_avi\_singles`, `kushagra\_singles`, `largetest\_doubles`, `gbaaddy`, `testlarge\_short`, `mahadevpura\_singles`) (8) | ~50 GB |
184: the *earliest* mtime across all known copies (C: working copy + **F**: archive ? robust to copy-resets;
185: a restored C: copy shows today, the **F**: archive keeps the real date). Fixes the `\_proxy` inconsistency
194: set vs the wider archive. Source = Google Drive ? bucket from a cloud box (not the local box).
259:- `docs/archives/past\_projects/DECOUPLED\_COMPUTE/` ? the cloud-engine design lineage.
===
1:# archives/roora-vendor/ ? Roora annotation-vendor engagement (archived 2026-06-21)
3:The **Roora data-labeling vendor** engagement docs, isolated from the live data-layer methodology (owner
9:- `ROORA\_LABELING\_GUIDELINES.md` ? the v8 vendor-facing labeling brief / annotation spec sent to Roora.
10:- `GOLDEN\_SET\_VENDORING.md` ? the vendor bake-off + Roora selection rationale (ADR-006).
13:> layer) because they're Roora-branded/vendor-facing (owner call, 2026-06-21). The reusable, vendor-neutral

36. user (2026-07-02T02:50:55.296Z)

```
1      // In-page annotation panel ? the browser analogue of the VLC dialog. Rendered into an
2      // OPEN Shadow DOM so the host page's CSS can't bleed in (and vice-versa) while still
3      // being reachable by automation/devtools ("closed" is not a real security boundary).
4      // Every user-facing string goes through the i18n shim t() (keys-not-prose);
reason/sport
5      // VALUES stay canonical (written to the CSV) ? only their display labels are
translated.
6
7      import { Annotator, SPORTS, REASON_OPTIONS, type Msg } from "../state/annotator";
8      import type { DirectVideoHandler } from "../video/directVideo";
9      import type { VideoIdentity } from "../persist/store";
10     import { VERSION } from "../version";
11     import { t, getLocale, setLocale, SUPPORTED_LOCALES, LOCALE_LABELS, type Locale } from
"../i18n";
12
13     export interface PanelDeps {
14       annotator: Annotator;
15       video: DirectVideoHandler;
16       identity: VideoIdentity;
17       download: (csv: string) => void;
18       onLocaleChange?: (locale: Locale) => void; // optional: persist the choice (content
script wires this)
19     }
20
21     export interface PanelHandle {
22       toggle(): void;
23       destroy(): void;
24     }
25
26     // mm:ss.mmm for display (matches the Lua fmt_clock).
27     function fmtClock(s: number | null): string {
```



```

28     if (s == null) return "--:--";
29     if (s < 0) s = 0;
30     const m = Math.floor(s / 60);
31     const sec = s - m * 60;
32     return `${m}:${sec.toFixed(3).padStart(6, "0")}`;
33 }
34
35 function esc(t: string): string {
36     return t.replace(/&/g, "&amp;").replace(/</g, "&lt;").replace(/>/g, "&gt;");
37 }
38
39 // Translate a Msg, mapping canonical reason/sport params to their display labels.
40 function tMsg(m: Msg): string {
41     let p = m.params;
42     if (p && (typeof p.reason === "string" || typeof p.sport === "string")) {
43         p = { ...p };
44         if (typeof p.reason === "string") p.reason = t("reason." + p.reason);
45         if (typeof p.sport === "string") p.sport = t("sport." + p.sport);
46     }
47     return t(m.key, p);
48 }
49
50 const STYLE = `
51 :host { all: initial; }
52 * { box-sizing: border-box; font-family: system-ui, "Noto Sans", "Noto Sans Devanagari",
53     "Noto Sans Kannada", "Noto Sans Telugu", "Nirmala UI", sans-serif; }
54 .card {
55     position: fixed; top: 16px; right: 16px; width: 340px; z-index: 2147483647;
56     background: #1e1f22; color: #e6e6e6; border: 1px solid #444; border-radius: 8px;
57     box-shadow: 0 8px 28px rgba(0,0,0,.5); font-size: 12px; user-select: none;
58 }
59 .hdr { display: flex; align-items: center; gap: 6px; padding: 7px 9px; cursor: move;
60     background: #2b2d31; border-radius: 8px 8px 0 0; }
61 .hdr .t { font-weight: 600; flex: 1; }
62 .hdr button { cursor: pointer; }
63 .body { padding: 9px; display: flex; flex-direction: column; gap: 7px; }
64 .row { display: flex; gap: 6px; align-items: center; }
65 .row > * { min-width: 0; }
66 label { color: #aab; white-space: nowrap; }
67 input, select, button {
68     font-size: 12px; background: #2b2d31; color: #e6e6e6; border: 1px solid #555;
69     border-radius: 5px; padding: 4px 6px;
70 }
71 input { width: 100%; }
72 button { cursor: pointer; background: #3a3d44; }
73 button:hover { background: #474b54; }
74 button.primary { background: #2f6f4f; border-color: #3c8a63; }
75 button.primary:hover { background: #38805c; }
76 .grid2 { display: grid; grid-template-columns: 1fr 1fr; gap: 6px; }
77 .status { background: #15161a; border: 1px solid #333; border-radius: 5px; padding: 6px;
78     line-height: 1.45; min-height: 52px; color: #cfd3da; }
79 .list { background: #15161a; border: 1px solid #333; border-radius: 5px; height: 122px;
80     overflow-y: auto; }
81 .list .item { padding: 3px 6px; cursor: pointer; border-bottom: 1px solid #232427;
82     white-space: nowrap; }
83 .list .item:hover { background: #25262b; }
84 .list .item.sel { background: #2f4a6f; }
85 .help { background: #15161a; border: 1px solid #333; border-radius: 5px; padding: 7px;
86     line-height: 1.5; max-height: 260px; overflow-y: auto; color: #cfd3da; }
87 .mini { font-size: 11px; color: #8b90a0; }
88 `;
89
90 function h<K extends keyof HTMLElementTagNameMap>(
91     tag: K,
92     props: Partial<HTMLElementTagNameMap[K]> & { class?: string } = {},

```

```

93     ...kids: (Node | string)[]
94 ): HTMLElementTagNameMap[K] {
95     const e = document.createElement(tag);
96     for (const [k, v] of Object.entries(props)) {
97         if (k === "class") e.className = v as string;
98         else (e as any)[k] = v;
99     }
100     for (const kid of kids) e.append(kid as any);
101     return e;
102 }
103
104 export function mountPanel(deps: PanelDeps): PanelHandle {
105     const { annotator: a, video, download } = deps;
106
107     const host = document.createElement("div");
108     host.style.all = "initial";
109     const root = host.attachShadow({ mode: "open" });
110     root.append(h("style", { textContent: STYLE }));
111
112     const card = h("div", { class: "card" });
113     root.append(card);
114
115     // ---- header (drag handle) ----
116     const titleEl = h("span", { class: "t" });
117     const helpBtn = h("button", { title: "Toggle help" });
118     const hideBtn = h("button", { textContent: "?", title: "Hide panel" });
119     const hdr = h("div", { class: "hdr" }, titleEl, helpBtn, hideBtn);
120     card.append(hdr);
121
122     const body = h("div", { class: "body" });
123     card.append(body);
124
125     // ---- language ----
126     const langLabel = h("label");
127     const langSel = h("select", { name: "language" }) as HTMLSelectElement;
128     for (const l of SUPPORTED_LOCALES) langSel.append(h("option", { value: l, textContent:
LOCALE_LABELS[l] }));
129     langSel.value = getLocale();
130     body.append(h("div", { class: "row" }, langLabel, langSel));
131
132     // ---- sport ----
133     const sportLabel = h("label");
134     const sportSel = h("select", { name: "sport" }) as HTMLSelectElement;
135     for (const s of SPORTS) sportSel.append(h("option", { value: s }));
136     sportSel.value = a.sport;
137     body.append(h("div", { class: "row" }, sportLabel, sportSel));
138
139     // ---- playback ----
140     const backBtn = h("button");
141     const playBtn = h("button");
142     const fwdBtn = h("button");
143     body.append(h("div", { class: "row" }, backBtn, playBtn, fwdBtn));
144
145     // ---- start / end ----
146     const startLabel = h("label");
147     const startIn = h("input", { name: "start" }) as HTMLInputElement;
148     const endLabel = h("label");
149     const endIn = h("input", { name: "end" }) as HTMLInputElement;
150     body.append(h("div", { class: "row" }, startLabel, startIn, endLabel, endIn));
151
152     // ---- next # / shots ----
153     const nextLabel = h("label");
154     const nextIn = h("input", { name: "next" }) as HTMLInputElement;
155     const shotsLabel = h("label");
156     const shotsIn = h("input", { name: "shots" }) as HTMLInputElement;

```

```

157     body.append(h("div", { class: "row" }, nextLabel, nextIn, shotsLabel, shotsIn));
158
159     // ---- reason ----
160     const reasonLabel = h("label");
161     const reasonSel = h("select", { name: "reason" }) as HTMLSelectElement;
162     for (const r of REASON_OPTIONS) reasonSel.append(h("option", { value: r }));
163     body.append(h("div", { class: "row" }, reasonLabel, reasonSel));
164
165     // ---- mark / save ----
166     const markStartBtn = h("button");
167     const markEndBtn = h("button");
168     const saveBtn = h("button", { class: "primary" });
169     body.append(h("div", { class: "grid2" }, markStartBtn, markEndBtn));
170     body.append(h("div", { class: "row" }, saveBtn));
171
172     // ---- status ----
173     const status = h("div", { class: "status" });
174     body.append(status);
175
176     // ---- help (hidden by default) ----
177     const help = h("div", { class: "help" });
178     help.style.display = "none";
179     body.append(help);
180
181     // ---- recent list ----
182     const recentLabel = h("div", { class: "mini" });
183     body.append(recentLabel);
184     const list = h("div", { class: "list" });
185     body.append(list);
186
187     // ---- actions ----
188     const editBtn = h("button");
189     const delBtn = h("button");
190     const undoBtn = h("button");
191     const refreshBtn = h("button");
192     const dlBtn = h("button");
193     body.append(h("div", { class: "grid2" }, editBtn, delBtn));
194     body.append(h("div", { class: "grid2" }, undoBtn, refreshBtn));
195     body.append(h("div", { class: "row" }, dlBtn));
196
197     // ---- state ----
198     let selectedN: number | null = null;
199     let lastMsg: Msg = { key: "status.ready" };
200
201     function setStatus(m: Msg) {
202         lastMsg = m;
203         render();
204     }
205
206     // Language-only text (static labels, option labels, placeholders, fixed buttons,
help).
207     // Re-applied on mount and whenever the locale changes.
208     function applyI18n() {
209         titleEl.textContent = `? ${t("panel.title")} v${VERSION}`;
210         langLabel.textContent = t("label.language");
211         sportLabel.textContent = t("label.sport");
212         for (const o of Array.from(sportSel.options)) o.textContent = t("sport." + o.value);
213         backBtn.textContent = t("btn.back5");
214         playBtn.textContent = t("btn.playPause");
215         fwdBtn.textContent = t("btn.fwd5");
216         startLabel.textContent = t("label.start");
217         endLabel.textContent = t("label.end");
218         nextLabel.textContent = t("label.next");
219         shotsLabel.textContent = t("label.shots");
220         startIn.placeholder = t("ph.start");

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```

221     endIn.placeholder = t("ph.end");
222     nextIn.placeholder = t("ph.next");
223     shotsIn.placeholder = t("ph.shots");
224     reasonLabel.textContent = t("label.reason");
225     for (const o of Array.from(reasonSel.options)) o.textContent = t("reason." +
o.value);
226     editBtn.textContent = t("btn.edit");
227     delBtn.textContent = t("btn.delete");
228     refreshBtn.textContent = t("btn.refresh");
229     dlBtn.textContent = t("btn.downloadCsv");
230     recentLabel.textContent = t("label.recent");
231     help.innerHTML = t("help.html");
232 }
233
234 function render() {
235     // fields
236     startIn.value = a.startField;
237     endIn.value = a.endField;
238     nextIn.value = a.nextField;
239     shotsIn.value = a.shotsField;
240     reasonSel.value = a.reason;
241     sportSel.value = a.sport;
242     langSel.value = getLocale();
243     // state-dependent labels
244     markStartBtn.textContent = tMsg(a.markStartLabel());
245     markEndBtn.textContent = tMsg(a.markEndLabel());
246     saveBtn.textContent = tMsg(a.saveLabel());
247     undoBtn.textContent = tMsg(a.undoLabel());
248     helpBtn.textContent = help.style.display === "none" ? t("btn.help") :
t("btn.hideHelp");
249     // status
250     const editing = a.mode === "edit" && a.editIndex !== null ? a.rows[a.editIndex] :
null;
251     const last = a.lastRow();
252     const lines = [
253         tMsg(lastMsg),
254         editing ? t("status.modeEdit", { n: editing.n }) : t("status.modeNew"),
255     ];
256     if (last) {
257         lines.push(
258             tMsg({
259                 key: "status.lastRow",
260                 params: { n: last.n, s: fmtClock(last.s), e: fmtClock(last.e), reason:
last.reason },
261             })
262         );
263     }
264     lines.push(t("status.footer", { now: fmtClock(video.now()), count: a.rows.length
}));
265     status.innerHTML =
266         lines.map(esc).join("<br>") +
267         `<br><span class="mini">${esc(t("status.csvLine", { key: deps.identity.key
})))}</span>`;
268     // list
269     list.replaceChildren();
270     for (const r of a.rows) {
271         const shotsTxt = r.shots !== null ? ` ${r.shots} ${t("ph.shots")}` : "";
272         const item = h("div", {
273             class: "item" + (r.n === selectedN ? " sel" : ""),
274             textContent: `#${r.n} ${fmtClock(r.s)} ? ${fmtClock(r.e)} [${t("reason." +
r.reason)}, ${t("sport." + r.sport)}]${shotsTxt}`,
275         });
276         item.addEventListener("click", () => {
277             selectedN = r.n;
278             render();

```

```

279         });
280         list.append(item);
281     }
282 }
283
284 // ---- wiring ----
285 startIn.addEventListener("input", () => {
286     a.startField = startIn.value;
287     render();
288 });
289 endIn.addEventListener("input", () => {
290     a.endField = endIn.value;
291     render();
292 });
293 nextIn.addEventListener("input", () => {
294     a.nextField = nextIn.value;
295     render();
296 });
297 shotsIn.addEventListener("input", () => {
298     a.shotsField = shotsIn.value;
299 });
300 reasonSel.addEventListener("change", () => {
301     a.reason = reasonSel.value;
302 });
303 sportSel.addEventListener("change", () => {
304     a.sport = sportSel.value;
305 });
306 langSel.addEventListener("change", () => {
307     setLocale(langSel.value as Locale);
308     deps.onLocaleChange?.(getLocale());
309     applyI18n();
310     render();
311 });
312
313 backBtn.addEventListener("click", () =>
314     setStatus({ key: video.seekBy(-5) ? "status.seekBack" : "status.noVideoSeek" })
315 );
316 fwdBtn.addEventListener("click", () =>
317     setStatus({ key: video.seekBy(5) ? "status.seekFwd" : "status.noVideoSeek" })
318 );
319 playBtn.addEventListener("click", () => {
320     const st = video.playPause();
321     setStatus({ key: st === "playing" ? "status.resumed" : st === "paused" ?
"status.paused" : "status.noVideoLoaded" });
322 });
323
324 markStartBtn.addEventListener("click", () => setStatus(a.markStart(video.now())));
325 markEndBtn.addEventListener("click", () => setStatus(a.markEnd(video.now())));
326 saveBtn.addEventListener("click", () => {
327     const res = a.saveRally();
328     if (res.ok) download(a.toCSV()); // download the full CSV on each save (per design)
329     setStatus(res);
330 });
331
332 editBtn.addEventListener("click", () => setStatus(a.editSelected(selectedN)));
333 delBtn.addEventListener("click", () => {
334     const res = a.deleteSelected(selectedN);
335     if (res.ok) selectedN = null;
336     setStatus(res);
337 });
338 undoBtn.addEventListener("click", () => setStatus(a.undoLast()));
339 refreshBtn.addEventListener("click", () => {
340     video.findActive();
341     setStatus({ key: video.hasVideo() ? "status.refreshedVideo" : "status.noVideoFound"
});

```

```

342     });
343     dlBtn.addEventListener("click", () => {
344         download(a.toCSV());
345         setStatus({ key: "status.downloaded", params: { count: a.rows.length } });
346     });
347
348     helpBtn.addEventListener("click", () => {
349         help.style.display = help.style.display === "none" ? "block" : "none";
350         render();
351     });
352     hideBtn.addEventListener("click", () => api.toggle());
353
354     // ---- dragging ----
355     let drag: { x: number; y: number; left: number; top: number } | null = null;
356     hdr.addEventListener("mousedown", (e) => {
357         const rect = card.getBoundingClientRect();
358         card.style.right = "auto";
359         card.style.left = rect.left + "px";
360         card.style.top = rect.top + "px";
361         drag = { x: e.clientX, y: e.clientY, left: rect.left, top: rect.top };
362         e.preventDefault();
363     });
364     const onMove = (e: MouseEvent) => {
365         if (!drag) return;
366         card.style.left = drag.left + (e.clientX - drag.x) + "px";
367         card.style.top = drag.top + (e.clientY - drag.y) + "px";
368     };
369     const onUp = () => {
370         drag = null;
371     };
372     window.addEventListener("mousemove", onMove);
373     window.addEventListener("mouseup", onUp);
374
375     // ---- live clock ----
376     const clock = window.setInterval(render, 333);
377
378     // ---- fullscreen re-parent (so the panel survives fullscreen video; desktop) ----
379     const onFs = () => {
380         const fe = document.fullscreenElement;
381         if (fe && !fe.contains(host)) fe.append(host);
382         else if (!fe && host.parentElement !== document.body) document.body.append(host);
383     };
384     document.addEventListener("fullscreenchange", onFs);
385
386     document.body.append(host);
387     applyI18n();
388     render();
389
390     const api: PanelHandle = {
391         toggle() {
392             host.style.display = host.style.display === "none" ? "" : "none";
393         },
394         destroy() {
395             window.clearInterval(clock);
396             window.removeEventListener("mousemove", onMove);
397             window.removeEventListener("mouseup", onUp);
398             document.removeEventListener("fullscreenchange", onFs);
399             host.remove();
400         },
401     };
402     return api;
403 }
404

```

```

1      // i18n ? a dependency-free t() shim over a shared common.json catalog, mirroring
2      // Khelsutra's marketing-site shim (avidullu/khelsutra `site/public/i18n.js`). Keys,
never
3      // prose; always fall back to `en`; a missing key degrades to en, then to the raw key
4      // (which the "every used key exists in en" test forbids in practice).
5      //
6      // rally-annotator's UI strings are deliberately PLURAL-FREE (counts are phrased
neutrally,
7      // e.g. "Rallies: {count}"), so the shim is interpolation-only ? no Intl.PluralRules and
8      // none of Spanish's 3-form-plural wrinkle. See docs/LOCALIZATION.md.
9
10     import en from "./locales/en/common.json";
11     import kn from "./locales/kn/common.json";
12     import hi from "./locales/hi/common.json";
13     import te from "./locales/te/common.json";
14     import es from "./locales/es/common.json";
15     import da from "./locales/da/common.json";
16     import id from "./locales/id/common.json";
17
18     export const SUPPORTED_LOCALES = ["en", "kn", "hi", "es", "da", "id", "te"] as const;
19     export type Locale = (typeof SUPPORTED_LOCALES)[number];
20     export const DEFAULT_LOCALE: Locale = "en";
21
22     // Native-script display names for the language selector.
23     export const LOCALE_LABELS: Record<Locale, string> = {
24         en: "English",
25         kn: "?????",
26         hi: "?????",
27         es: "Español",
28         da: "Dansk",
29         id: "Bahasa Indonesia",
30         te: "?????",
31     };
32
33     // Scripts that need a bundled OFL Noto font (Latin locales render with the system
stack).
34     export const NON_LATIN: ReadonlySet<Locale> = new Set<Locale>(["kn", "hi", "te"]);
35
36     type Catalog = Record<string, string>;
37
38     // Registered catalogs. `en` is the source of truth; the six translated catalogs are
39     // machine drafts pending native review (see each file's _meta). An unregistered locale,
40     // or any key missing from a catalog, falls back to en.
41     const C = (j: unknown) => j as unknown as Catalog;
42     const CATALOGS: Partial<Record<Locale, Catalog>> = {
43         en: C(en),
44         kn: C(kn),
45         hi: C(hi),
46         te: C(te),
47         es: C(es),
48         da: C(da),
49         id: C(id),
50     };
51
52     export function registerCatalog(locale: Locale, catalog: Catalog): void {
53         CATALOGS[locale] = catalog;
54     }
55
56     let current: Locale = DEFAULT_LOCALE;
57     export function getLocale(): Locale {
58         return current;
59     }
60     export function setLocale(locale: Locale): void {
61         current = (SUPPORTED_LOCALES as readonly string[]).includes(locale) ? locale :
DEFAULT_LOCALE;

```

```

62     }
63
64     function toSupported(code?: string | null): Locale | null {
65         if (!code) return null;
66         const base = code.toLowerCase().split("-")[0];
67         return (SUPPORTED_LOCALES as readonly string[]).includes(base) ? (base as Locale) :
null;
68     }
69
70     // Negotiation order (shared with Khelsutra): a stored/explicit preference wins, then
the
71     // browser's languages, then `en`. Pure ? safe to call with no arguments outside a
browser.
72     export function negotiateLocale(stored?: string | null, navLangs: readonly string[] =
[]): Locale {
73         const fromStored = toSupported(stored);
74         if (fromStored) return fromStored;
75         for (const l of navLangs) {
76             const m = toSupported(l);
77             if (m) return m;
78         }
79         return DEFAULT_LOCALE;
80     }
81
82     function interpolate(tpl: string, vars?: Record<string, string | number>): string {
83         if (!vars) return tpl;
84         return tpl.replace(/\{(\w+)\}/g, (_m, k: string) => (k in vars ? String(vars[k]) :
`{${k}}`));
85     }
86
87     // t(key, vars?, locale?) ? resolve against the locale, then en, then the raw key.
88     export function t(
89         key: string,
90         vars?: Record<string, string | number>,
91         locale: Locale = current
92     ): string {
93         const tpl = CATALOGS[locale]?.[key] ?? CATALOGS.en?.[key] ?? key;
94         return interpolate(tpl, vars);
95     }
96
97     // ---- introspection used by the key-parity / key-presence tests ----
98     export function registeredLocales(): Locale[] {
99         return Object.keys(CATALOGS) as Locale[];
100     }
101     // Catalog keys excluding the `_meta` bookkeeping entry.
102     export function catalogKeys(locale: Locale): string[] {
103         return Object.keys(CATALOGS[locale] ?? {}).filter((k) => k !== "_meta");
104     }
105     export const EN_KEYS: string[] = catalogKeys("en");
106

```

38. user (2026-07-02T02:51:24.534Z)

```

===tests===
test/i18n.test.ts

```

39. user (2026-07-02T02:51:51.291Z)

```

88:local VERSION = "1.7.1"
end

local function lang_index(code)
    for i, c in ipairs(LOCALES) do if c == code then return i end end
    return 1
end

```



```

-- Persist the chosen language in a tiny config file (extensions have no settings API).
local function lang_config_path()
    local home = os.getenv("USERPROFILE") or os.getenv("HOME") or "."
    local sep = home:find("\\") and "\\" or "/"
    if home:sub(-1) ~= sep then home = home .. sep end
    return home .. ".rally_annotator_lang"
end
local function load_lang()
    local f = io.open(lang_config_path(), "r")
    if not f then return end
    local code = f:read("*l"); f:close()
    if code then code = code:gsub("%s+", "") end
    if code and STRINGS[code] then LANG = code end
end
local function save_lang()
    local f = io.open(lang_config_path(), "w")
    if f then f:write(LANG .. "\n"); f:close() end
end

-----
-- State
-----

local d                -- the single dialog (one per extension)
local w_status          -- status label widget (HTML/rich-text); also hosts help
local w_reason          -- ending_reason dropdown widget
local w_sport           -- sport dropdown widget
local w_start           -- start-time text input (editable seconds)
local w_end             -- end-time text input (editable seconds)
local w_list            -- recent-rallies list widget
local w_save            -- "Save Rally" / "Save changes" button (re-labeled by state)
local w_mark_start      -- "Mark START" button (re-labeled "Re-mark START (#N)" while editing)
local w_mark_end        -- "Mark END" button (re-labeled "Re-mark END (#N)" while editing)
local w_undo            -- "Undo last" button (re-labeled to show which row it removes)
local w_next            -- "Next rally #" text input (lets you resume numbering anywhere)
local w_shots           -- "Number of shots" text input (optional shots_count per rally)
local w_help_btn        -- the Help button (re-labeled "Hide help" when open)
local w_help            -- the help panel widget; nil when hidden (toggled via del_widget)
local w_lang            -- language-selector dropdown (en/kn/hi/te/es/da/id)

local rows = {}         -- in-memory rally rows: { n, s, e, reason, sport, [shots], [extra] }
local mode = "new"      -- "new" (marking a fresh rally) or "edit" (editing a row)
local edit_index = nil  -- index into rows when mode == "edit"
local out_path          -- resolved CSV path (set on activate; re-resolved by Refresh)
local out_path_is_fallback = false -- true when out_path is the home-dir fallback (no video
                                -- was resolvable when we last set it); see
adopt_current_video_csv

-----
-- Helpers (declared before the global callbacks that capture them)
-----

-- Current playback time in SECONDS, or nil if nothing is playing.

```

40. user (2026-07-02T02:52:04.002Z)

```

// Single source of truth for the version, mirrored into the panel title and the
// manifest (via package.json). Same anti-drift idea as the Lua VERSION constant.
export const VERSION = "0.1.0";

```

Changelog

v1.7.1 ? 2026-06-22

```

- **Completes the localization:** the dialog's **status messages** and the full in-dialog **HELP
/
  ending-reason guide** are now localized into all 7 languages too (en/hi/kn/te/es/da/id) ? so

```

the
 entire dialog is translated, not just the chrome. Status lines show localized reason/sport labels
 while the CSV keeps **canonical English** values (mapped via the dropdown id). The machine-draft catalogs remain **pending native-speaker review**.
 - The `STRINGS` table now carries 81 keys/locale; the headless suite's key-parity gate spans them all.
 88 assertions green (`lua5.1 test/dialog\_test.lua`); `luac5.1 -p` clean.

v1.7 ? 2026-06-22

```
50:| Persist | `src/persist/store.ts` | `chrome.storage.local` live store + per-video identity + CSV filename |
140:- **LD-11 ? Single-file CSV mode.** `showSaveFilePicker` + a persisted `FileSystemFileHandle` to rewrite
```

41. user (2026-07-02T02:52:11.058Z)

```
55:### LD-i8 ? Language selection, persisted
56:Each UI has a language selector. Web: persisted in `chrome.storage.local` (seeded from the browser
57:language; default `en`). VLC: persisted in a tiny config file (extensions have no settings API).
70:- `panel.ts`: every label/button/help/status via `t()`; add a language `` (persisted); inject the
79: Persist the choice in a config file.
102: rebuilds the dialog + persists the choice + **id-based canonical reason/sport** so the CSV is unchanged;
===
129:**Decision.** The panel UI is localized into Khelsutra's locales **+ Telugu** (`en, kn, hi, es, da, id,
133:English (LD-3 byte-compatibility is preserved across all languages). Full design + locked decisions:
```

42. user (2026-07-02T02:52:45.810Z)

```
import { defineConfig } from "vitest/config";

export default defineConfig({
  test: {
    environment: "node",
    include: ["test/**/*.test.ts"],
    coverage: {
      provider: "v8",
      reporter: ["text-summary", "text", "lcov"],
      include: ["src/**/*.ts"],
      exclude: ["src/messages.ts"], // type-only
      thresholds: {
        // Enforced floors ? the build fails below these (currently ~98% stmts/lines/funcs,
        // ~86% branches; floors sit just under, as a stable guard). Pure logic
        // (csv/annotator/store) is ~100%; the few uncovered branches are defensive
        // fallbacks, and the real browser behavior is proven by the Playwright E2E.
        statements: 95,
        branches: 82,
        functions: 95,
        lines: 95,
      },
    },
  },
});
```

43. user (2026-07-02T02:52:53.994Z)

```
7:// EACH supported locale, switch the panel's language, prove the UI renders in that language,
```

```

9:// (English) reason/sport values. Each locale's session is screen-recorded to
10:// e2e-results/screencasts/<locale>.webm ? the downloadable per-language proof.
20:const LOCALES_DIR = path.resolve(here, "..", "src", "i18n", "locales");
31:// Mirror of the runtime t(): locale -> en -> raw key, with {var} interpolation.
32:function tt(locale: string, key: string, vars?: Record<string, string | number>): string {
33:  const tpl = CATS[locale]?.[key] ?? CATS.en[key] ?? key;
49:for (const locale of SUPPORTED_LOCALES) {
50:  test(`panel works and renders in ${locale} (${LOCALE_LABELS[locale]})`, async () => {
51:    const T = (key: string, vars?: Record<string, string | number>) => tt(locale, key, vars);
53:    // Fresh per-locale recording dir so a single recording is unambiguous (robust across
reruns).
54:    const ldir = path.join(VIDEO_DIR, locale);
78:    await expect(page.locator(".hdr .t")).toContainText("Rally Annotator"); // brand
constant
80:    // Switch to this locale and PROVE the UI renders in that language.
81:    await page.locator("select[name=language]").selectOption(locale);
82:    await expect(page.getByRole("button", { name: T("btn.markStart") })).toBeVisible();
90:    await expect(page.getByPlaceholder(T("ph.start"))).toHaveValue("1.000");
91:    await expect(page.getByPlaceholder(T("ph.end"))).toHaveValue("3.500");
97:    await expect(page.locator(".item")).toHaveCount(1);
98:    await expect(page.locator(".item").first()).toContainText("#1");
99:    await expect(page.locator(".item").first()).toContainText(T("reason.winner"));
101:    // Authoritative proof: persisted with CANONICAL english reason/sport regardless of UI
language.
107:    expect(Array.isArray(rows)).toBe(true);
109:    expect(list).toHaveLength(1);
110:    expect(list[0]).toMatchObject({ n: 1, reason: "winner", sport: "badminton", shots:
"12" });
111:    expect(Number(list[0].e)).toBeGreaterThan(Number(list[0].s));
121:    copyFileSync(src, path.join(SCREENCAST_DIR, `${locale}.webm`));

```

44. user (2026-07-02T02:54:34.651Z)

Structured output provided successfully